MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COUNTER, BINARY, DIGITAL, KEY CODE FHA

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope This document establishes the requirements and test procedures for the acquisition of an electronic binary counter module

* 1.2 Classification This specification covers two classes of modules

Class I and class II as specified in MIL-STD-1389

2 APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 Design Requirements for Standard Electronic Modules

MIL-STD 1666 Test Equipment for the Standard Electronic Modules Program

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence

3. REQUIREMENTS

3.1 General requirements The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787

Beneficial comments (recommendations, additions, deletions and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: PME 117-[illegible], Washington, DC 20363-5100 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document

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* 3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.25 failures per million operating hours for class I or 0 35 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389 When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6 4).

* 3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I . Class I modules shall conform to the electrical requirements of class I only, and class II modules shall comform to the electrical requirements of class I and class II (see 6 4)

3.5 Module configuration requirements The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5 1 Key code. The module key code shall be FHA.

3.5.2 Size. The module size shall be 1A.

3 5 3 Weight The module shall have a maximum weight of 35 grams

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40

3 5 5 Fin position The module shall have an insertion and extraction and heat dissipation fin on increment A

3 5 6 Depth The module shall have no additional depth requirements

3 5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes

* 3 5 8 Marking Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/70-1 for class I and M28787/70-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier)

3 6 Circuit pin configuration The function diagram and pin assignments shall conform to figure 1

3 7 Acceptance tests All production modules shall be tested in accordance with footnotes 1 and 2 of table I

3 8 End-of-life limits Modules which have been plugged into an operating system or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6 2)

3 9 Power supply voltage The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0 5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5 5 V dc ±1 percent

3.10 Functional tests. All modules shall be subjected to functional tests which conform to the requirements specified in 3 10 1 through 3 10.3

3.10 1 Test patterns. A set of unique input patterns shall be generated capable of detecting all nonredundant failures in the module logic network The test patterns shall be generated by assuming "stuck at H state" and "stuck at L state" failures in each of the basic logic functions and applying input patterns to detect these failures. The unique test patterns necessary to satisfy these requirements shall be submitted with the qualification samples If the tests of tables I and IX-XII satisfy or exceed these unique test patterns, no additional tests shall be added, otherwise, the needed tests shall be added to tables IX-XII "Stuck at H state" indicates a failure resulting in a steady high-level output "Stuck at L state" indicates a failure resulting in a steady low-level output Prior to any

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change in module design which would necessitate a different set of test patterns to detect a "stuck at H state" or a "stuck at L state", the vendor shall be required to submit to the acquiring activity and SEMP Quality Assurance Activity the proposed new set of test patterns for approval.

3.10.2 Input requirements. The following input voltage levels shall be used for test patterns:

Low level - Input voltage shall be 0.8 V ±1%.
High level - Input voltage shall be 2 V ±1%.

3.10.3 Output requirements. The output requirements for the test patterns shall be as follows:

Low level - Output voltage shall be 0 V minimum, 0.5 V maximum.
High level - Output voltage shall be 2.5 V minimum, V_{CC} maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through EE shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions:

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through E in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through DD shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition DD, shall be performed using the initial limits at 25°C ±5°C ambient.

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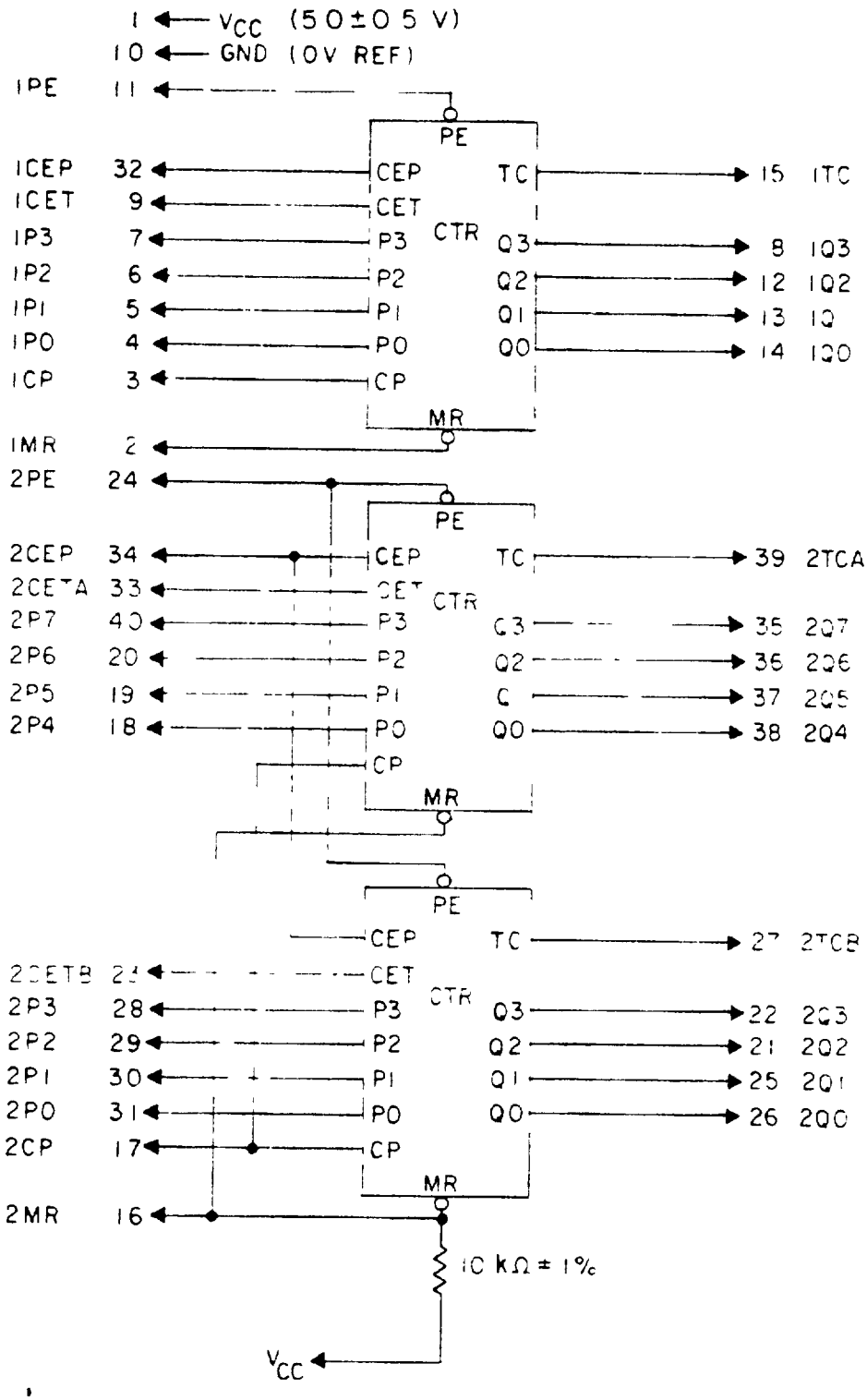


FIGURE 1 FHA module function diagram, binary counter

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*TABLE I Module electrical requirements

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B15	V_{OUT} (low-loaded)	IV	S	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	V
C1-C9, C12-C21	Input breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
C10,C11, C22,C23	Input breakdown current	V	S	0	2	0	2	0	2	0	2	0	2	mA
D1-D15	Output short circuit current	VI	X	-40	-100	-40	-100	-40	-100	-40	-100	-40	-100	mA
E1-E15	Output leakage current	VII	X	0	1	0	1	0	1	0	1	0	1	mA
F1 F23	Input clamp diode voltage	VIII	X	-0.4	-1.2	-0.4	-1 2	-0.4	-1.2	-0 4	-1 2	-0.4	-1.2	V
G1-G39	Functional	IX	X	← See table IX →										
H1-H15	Output voltage, low level	X	X	0	500	0	500	0	500	0	500	0	500	mV
J1 J15	Output voltage, high level	X	X	2.5	4 5	2.5	4 5	2.5	4 5	2.5	4 5	2.5	4 5	V
K1,K3,K5	Input threshold, low level	XI	X	← See table XI →										
K2,K4,K6	Input threshold, high level	XI	X	← See table XI →										
L1-L3, L12,L17	Input current, low level	XII	X	-0 1	-4.4	-0.1	-4.4	-0.1	-4 4	-0.1	-4.4	-0.1	4.4	mA
L4-L7,L9, L13-L16, L18-L21	Input current, low level	XII	X	-0 1	-2.2	-0 1	-2.2	-0.1	-2.2	-0.1	-2.2	-0 1	-2.2	mA
L8	Input current, low level	XII	X	-0.1	-5.5	-0 1	-5 5	-0 1	-5 5	-0.1	-5.5	-0.1	-5 5	mA
L10,L11	Input current, low level	XII	X	-0.1	-8.8	-0 1	-8.8	-0.1	8.8	-0.1	-8.8	-0.1	-8.8	mA
L22	Input current, low level	XII	X	-0.1	-11.0	-0.1	-11 0	-0.1	-11.0	-0.1	-11.0	-0.1	-11.0	mA
L23	Input current, low level	XII	X	-0 1	-5 1	-0 1	-5.1	-0.1	-5 1	-0.1	-5.1	-0.1	-5.1	mA

See footnotes at end of table.

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*TABLE I Module electrical requirements Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M1-M3, M12,M17	Input current, high level	XIII	X	-1	100	-1	100	-1	100	-1	100	-1	100	µA
M4-M7 M9, M13-M16 M18-M21	Input current, high level	XIII	X	-1	50	-1	50	-1	50	-1	50	-1	50	µA
M8	Input current, high level	XIII	X	-1	125	-1	125	-1	125	-1	125	-1	125	µA
M10,M11	Input current, high level	XIII	X	-1	200	-1	200	-1	200	-1	200	-1	200	µA
M22	Input current, high level	XIII	X	-1	250	-1	250	-1	250	-1	250	-1	250	µA
M23	Input current, high level	XIII	X	-1	-315	-1	-315	-1	-315	-1	-315	-1	-315	µA
	Power supply current, low level outputs	XIV	X	50	382	50	382	50	382	50	382	50	382	mA
M1-P1	Transition time (t_{TLH}), C_L = 50 pF	XV	S	4	17	4	18	4	18	4	19	4	19	ns
M16-P16	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	6	27	6	30	6	30	6	32	6	32	ns
M1-Q1	Transition time (t_{THL}), C_L = 50 pF	XV	S	2	9	2	10	2	10	2	12	2	12	ns
M16 Q16	Transition time (t_{THL}), C_L = 150 pF	XV	Q	2	11	2	13	2	13	2	15	2	15	ns
M1,R11	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	6	21	6	23	6	23	6	25	6	25	ns
M2-R2 M7 R7, M12 R12, M15 R15	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	4	13	4	15	4	15	4	17	4	17	ns
M16,R21, M26	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	6	29	6	31	6	31	6	33	6	33	ns
M17-R20, M22-R25, M27 R30	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	4	21	4	23	4	23	4	25	4	25	ns

e 4/ ...tes at end of table

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*TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S1,S6,S11	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	4	16	4	18	4	18	4	20	4	20	ns
S2-S5, S7-S10, S12-S15	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	5	18	5	20	5	20	5	22	5	22	ns
S16,S21, S26	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	4	24	4	26	4	26	4	28	4	28	ns
S17-S20, S22-S25, S27-S30	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	5	26	5	28	5	28	5	30	5	30	ns
T1-T5	Propagation delay time (t_{PHL}), C_L = 50 pF	XVI	S	5	25	5	27	5	27	5	29	5	29	ns
T16-T30	Propagation delay time (t_{PHL}), C_L = 150 pF	XVI	Q	5	33	5	35	5	35	5	37	5	37	ns
U1-U3	Propagation delay time (t_{PLH}), C_L = 50 pF	XVII	Q	3	13	3	15	3	15	3	17	3	17	ns
U4-U6	Propagation delay time (t_{PLH}), C_L = 150 pF	XVII	Q	3	19	3	21	3	21	3	23	3	23	ns
V1-V3	Propagation delay time (t_{PHL}), C_L = 50 pF	XVII	S	3	14	3	16	3	16	3	18	3	18	ns
V4-V6	Propagation delay time (t_{PHL}), C_L = 150 pF	XVII	Q	3	20	3	22	3	22	3	24	3	24	ns
W1,W3,W5	Maximum clock frequency and minimum pulse width (PW1)	XVIII	S	21	29	21	29	21	29	21	29	21	29	ns
W2,W4,W6	Maximum clock frequency and minimum pulse width (PW2)	XVIII	S	16	24	16	24	16	24	16	24	16	24	ns
X1-X6	Setup time (t_S) check (measuring t_C)	XIX	S	1550	1650	1550	1650	1550	1650	1550	1650	1550	1650	ns
Y1-Y6	Hold time (t_H) check (measuring t_C)	XIX	S	1550	1650	1550	1650	1550	1650	1550	1650	1550	1650	ns

See footnotes at end of table.

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*TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Z1-Z3	Setup and hold time check (t_{PLH})	XX	S	4	13	4	15	4	15	4	17	4	17	ns
Z4-Z15	Setup and hold time, low level, check (t_{PHL})	XX	S	5	18	5	20	5	20	5	22	5	22	ns
Z16-Z27	Setup and hold time, high level, check (t_{PLH})	XX	S	4	13	4	15	4	15	4	17	4	17	ns
AA1-AA15	Output crosstalk, low level	XXI	Q	0	1,250	0	1,250	0	1,250	0	1,250	0	1,250	mV
AA16-AA30	Output crosstalk, duration	XXI	Q	0	500	0	500	0	500	0	500	0	500	mV
BB1-BB12	Output crosstalk, high level	XXI	Q	0	400	0	400	0	400	0	400	0	400	mV
CC	Power supply transient amplitude	XXII	Q	0	400	0	400	0	400	0	400	0	400	mV
DD	Power supply filter capacitance	XXIII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
EE1,EE2	Input capacitance	XXIV	Q	5	20	5	20	5	20	5	20	5	20	pF
EE3,EE12 EE17	Input capacitance	XXIV	Q	5	40	5	40	5	40	5	40	5	40	pF
EE4-EE9, EE13 EE16, EE18-EE21	Input capacitance	XXIV	Q	5	30	5	30	5	30	5	30	5	30	pF
EE10,EE11, EE22,EE23	Input capacitance	XXIV	Q	5	60	5	60	5	60	5	60	5	60	pF

1/ "X" parameters shall be 100 percent tested during module acceptance test
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C
3/ "Q" parameters shall be tested during qualification testing only
4/ Unless otherwise specified, limits are for initial and end-of-life

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through DD shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through DD shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/70-1 supersedes part number M28787/70. The two parts are interchangeable.

* 6.4 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.5 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	2.096 max	2.096 max	mW
Input current			
Low level (V_{IL} = 0 V)			
2MR	-5 max	-5 max	mA
1PE, 1CET, 2CETA, 1CEP, 2CETB	-4.4 max	-4.4 max	mA
1CP	-5.5 max	-5.5 max	mA
2PE, 2CEP	-8.8 max	-8.8 max	mA
2CP	-11 max	-11 max	mA
All others	-0.2 max	-0.2 max	mA
High level (V_{IH} = 2.5 V)			
1PE, 1CET, 2CETA, 1CEP 2CETB	100 max	100 max	µA
1CP	125 max	125 max	µA
2PE, 2CEP	200 max	200 max	µA
2CP	250 max	250 max	µA
2MR	-315 max	-315 max	µA
All others	50 max	50 max	µA
Output drive current			
Low level (V_{OL} = 0.5 V)	20 max	20 max	mA
High level (V_{OH} = 2.5 V)	-1 max	-1 max	mA
Setup and hold time			
Data input setup time (t_S)			
High level	10 min	10 min	ns
Low level	10 min	10 min	ns
CET, CEP input setup time (t_S)			
High level	12 min	12 min	ns
Low level	12 min	12 min	ns

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	Class I	Class II	Unit
PE input setup time (t_S)			
High level	16 min	16 min	ns
Low level	16 min	16 min	ns
Hold time, all inputs (t_H)			
High level	0 min	0 min	ns
Low level	0 min	0 min	ns
Clock input			
Frequency (C_L = 50 pF)	45 max	45 max	MHz
Pulse width (high)	8 min	8 min	ns
Trigger edge	positive	positive	
Reset pulse width	13 min	13 min	ns
Reset recovery time	8 min	8 min	ns

*6 6 Module function. This module contains one completely independent 4-bit synchronous binary counter and two additional 4-bit synchronous binary counters that have common clock, reset, PE, and CEP lines.

6 6 1 Input to output relationship. The synchronous binary counter circuit has the input to output relationship of table II for positive logic. The FHA module contains three binary counters which are fully synchronous with the clock pulse. Each counter contains four master/slave flip-flops which are driven, in parallel, by the clock through a clock buffer. The master is inhibited from further change during the low-to-high transition of the clock. Data is transferred from the masters to the slaves and reflected at the outputs after the masters are inhibited. When the clock is high, the masters remain inhibited and the master/slave data path remains established. During the clock high-to-low transition, the slave is inhibited from further change, followed by the enabling of the masters for acceptance of new data from the counting logic or the parallel entry logic.

a. There are three control inputs: Parallel Enable (PE), Count Enable Parallel (CEP), and Count Enable Trickle (CET). These control inputs select the mode of operation as shown in table I. When the conditions for counting are satisfied, i.e., CEP, CET, and PE are high, the rising edge of the clock will change the counters to the state of the count sequence.

b. The counter can be synchronously preset from the four parallel inputs (P0-P3) when PE is low. When the PE and clock are low, each master of the flip-flops is connected to the appropriate parallel input (P0-P3) and the slaves (outputs) are steady in their previous state. When the clock goes high, this information is transferred to the slaves and reflected at the outputs. The PE overrides both count enables presetting the counter low.

c. Terminal count is high when the counter is at a terminal count (state 15) and the CET is high.

d. The asynchronous master reset (MR) overrides all other inputs resetting each output low.

e. The PE, P0-P3, CET, and CEP inputs are completely edge triggered, i.e., responding only to the low-to-high transition of the clock line.

f. Long synchronous counters are constructed by connecting the TC output of the first counter to the CEP input of all other counters and the TC output of each counter after the first to the CET input of the next counter.

6 6 2 Parallel outputs. Paralleling of outputs is not permissible.

6 6 3 False clocking. To avoid possible false clocking due to slow clock rise time, Schottky circuits should be used to drive the clock.

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6.7 Changes from previous issue The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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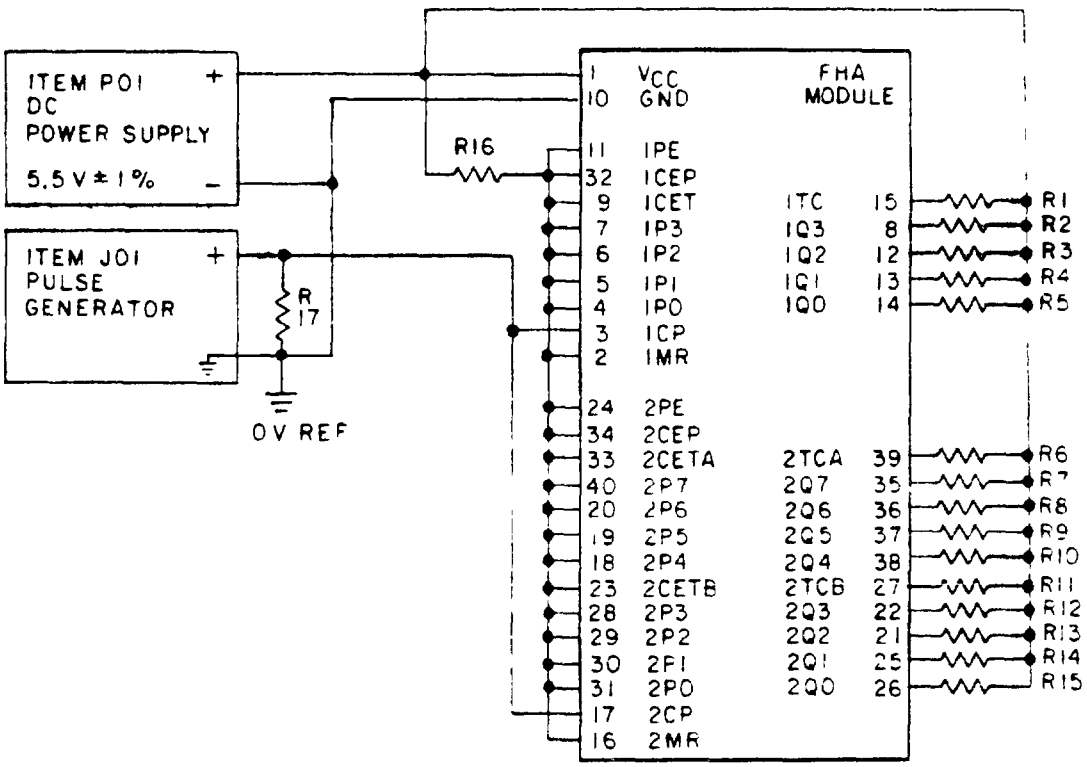
TABLE II Input to output relationship 1/ - 5/

Synchronous inputs before clock pulse							Required condition of asynchronous inputs	Outputs after clock pulse				
CEP	CET	PE	P3	P2	P1	P0	MR	TC	Q3	Q2	Q1	Q0
X	L	H	D3	D2	D1	D0	H	L	NC	NC	NC	NC
L	X	H	D3	D2	D1	D0	H	L	NC	NC	NC	NC
X	X	L	D3	D2	D1	D0	H	L	D3	D2	D1	D0
X	L	L	H	H	H	H	H	L	H	H	H	H
X	H	L	H	H	H	H	H	H	H	H	H	H
X	X	L	L	L	L	L	H	L	L	L	L	L
H	H	H	X	X	X	X	H	L	L	L	L	L
H	H	H	X	X	X	X	H	L	L	L	L	H
H	H	H	X	X	X	X	H	L	L	L	H	L
H	H	H	X	X	X	X	H	L	L	L	H	H
H	H	H	X	X	X	X	H	L	L	H	L	L
H	H	H	X	X	X	X	H	L	L	H	L	H
H	H	H	X	X	X	X	H	L	L	H	H	L
H	H	H	X	X	X	X	H	L	L	H	H	H
H	H	H	X	X	X	X	H	L	H	L	L	L
H	H	H	X	X	X	X	H	L	H	L	L	H
H	H	H	X	X	X	X	H	L	H	L	H	L
H	H	H	X	X	X	X	H	L	H	L	H	H
H	H	H	X	X	X	X	H	L	H	H	L	L
H	H	H	X	X	X	X	H	L	H	H	L	H
H	H	H	X	X	X	X	H	L	H	H	H	L
H	H	H	X	X	X	X	H	H	H	H	H	H
H	H	H	X	X	X	X	H	L	L	L	L	L

Asynchronous inputs		Required condition of synchronous inputs							Outputs				
MR	CET	CP	PE	CEP	P3	P2	P1	P0	TC	Q3	Q2	Q1	Q0
H	X	X	X	X	X	X	X	X	X	X	X	X	X
L	X	X	X	X	X	X	X	X	L	L	L	L	L
H	I	L	L	X	H	H	H	H	NC	NC	NC	NC	NC
H	I	H	L	X	H	H	H	H	NC	H	H	H	H
H	L	H	H	L	X	X	X	X	NC	NC	NC	NC	NC
H	L	H	H	H	X	X	X	X	NC	NC	NC	NC	NC
H	H	H	H	L	X	X	X	X	H	NC	NC	NC	NC
H	L	H	H	X	X	X	X	X	L	NC	NC	NC	NC
H	H	L	H	L	X	X	X	X	H	NC	NC	NC	NC
H	L	L	H	X	X	X	X	X	L	NC	NC	NC	NC

1/ NC = No change from previous state
2/ D0-D3 = Known input and output data (either high or low)
3/ X = Unspecified inputs or outputs (either high or low)
4/ H = High level.
5/ L = Low level.

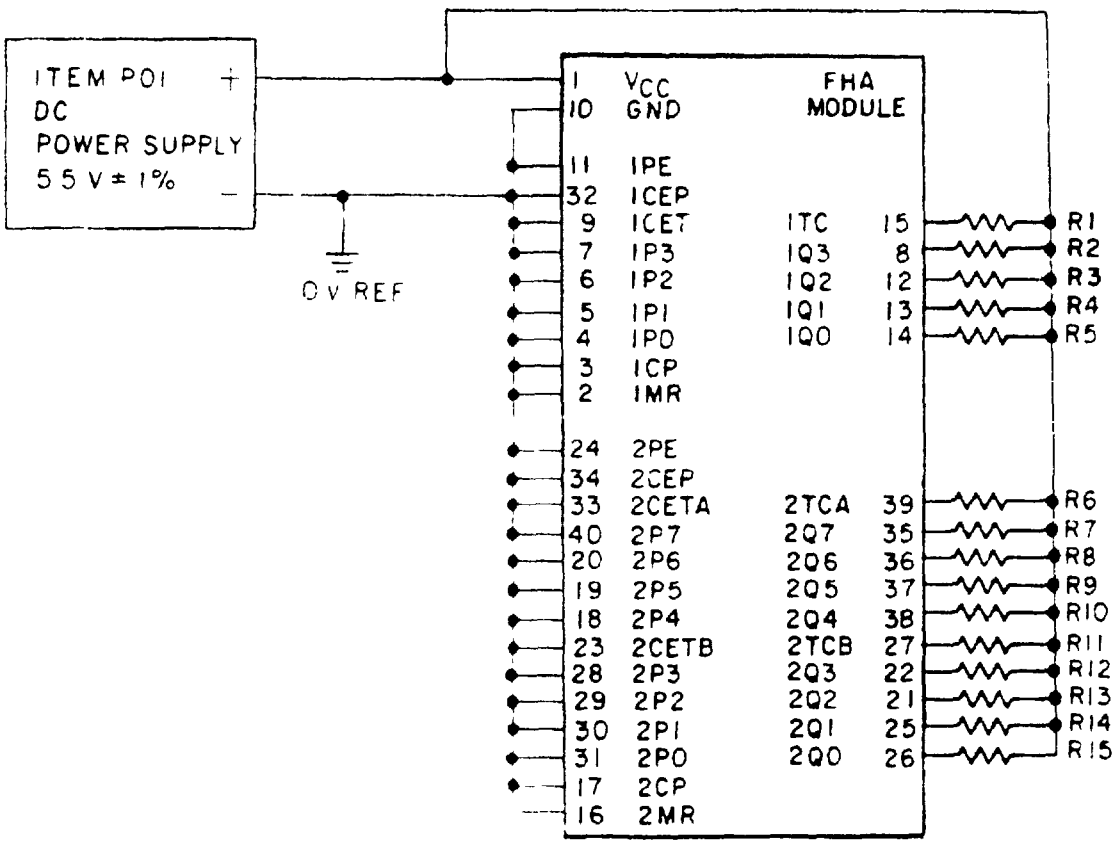
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NOTES
1. P1 = R6 = R11 = 249Ω ±1%, 1/4 W.
2. R2-R5 = R7-R10 = R12-R15 = 316Ω ±1%, 1/4 W.
3. R16 = 1 kΩ ±5%, 1/4 W
4. R17 = 51Ω ±5%, 1/4 W
5. Item J01 setup
   PW = 500 ±100 ns.
   PRR = 1 MHz ±20%.
   PA = 3.5 ±0.5 V.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.

FIGURE 2 Operating temperature and life test circuit

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NOTE R1-R15 = 240 Ω ±1 %, 1/4 W

FIGURE 3 Component temperature test circuit

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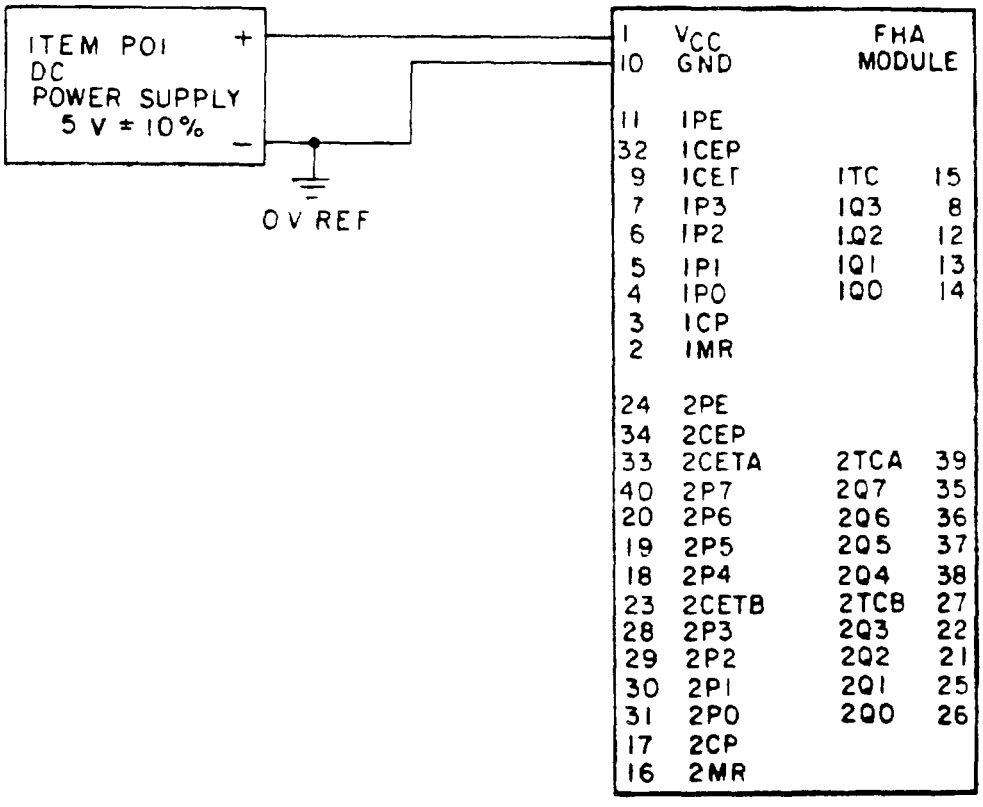


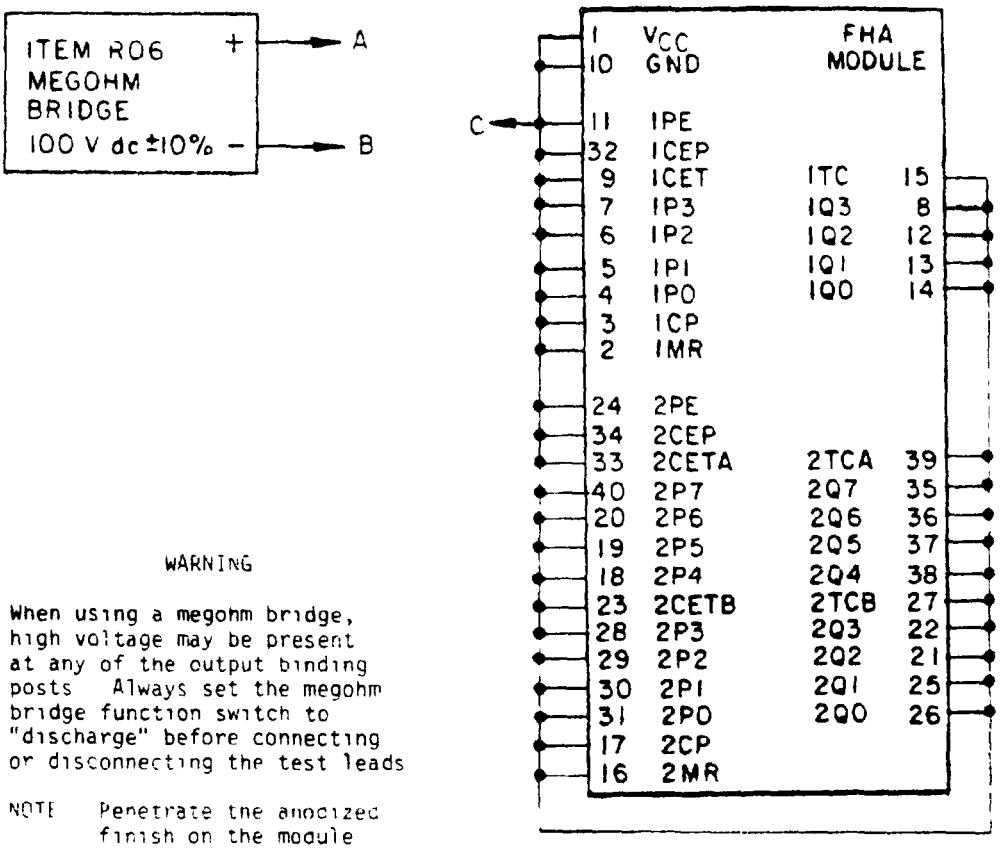
FIGURE 4 Vibration test circuit

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	5

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

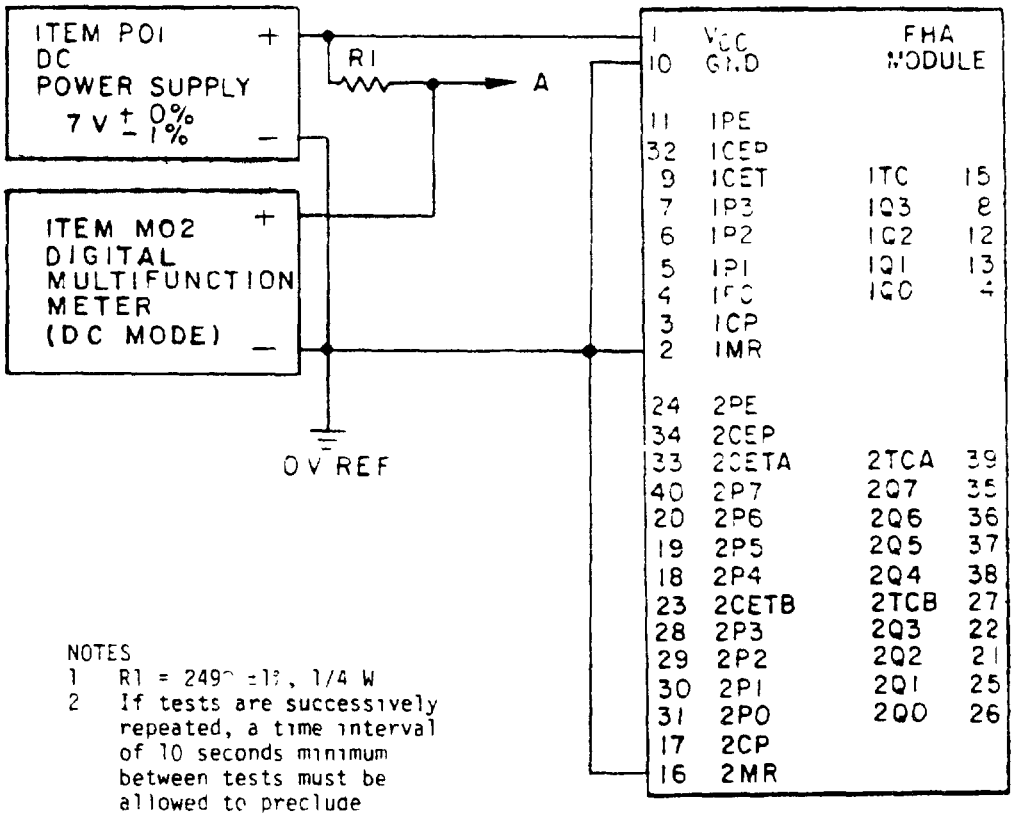
FIGURE 5 Isolation test circuit for table III

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TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
B1	1TC V_{OUT} (low loaded)	15	0.01	1.00	V	6
B2	1Q3 "	8	"	"	"	"
B3	1Q2 "	12	"	"	"	"
B4	1Q1 "	13	"	"	"	"
B5	1Q0 "	14	"	"	"	"
B6	2TCA "	39	"	"	"	"
B7	2Q7 "	35	"	"	"	"
B8	2Q6 "	36	"	"	"	"
B9	2Q5 "	37	"	"	"	"
B10	2Q4 "	38	"	"	"	"
B11	2TCB "	27	"	"	"	"
B12	2Q3 "	22	"	"	"	"
B13	2Q2 "	21	"	"	"	"
B14	2Q1 "	25	"	"	"	"
B15	2Q0 "	26	"	"	"	"

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NOTES
1 R1 = 249Ω ±1%, 1/4 W
2 If tests are successively repeated, a time interval of 10 seconds minimum between tests must be allowed to preclude detrimental effects due to module heating

* FIGURE 6 V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	1PE	Input breakdown current	11	2,3,4,5,6,7,9,16,17,18,19,20,23, 24,28,29,30,31,32,33,34,40	0	1	mA	7
C2	1CEP	"	32	2,3,4,5,6,7,9,11,16,17,18,19,20, 23,24,28,29,30,31,33,34,40	"	"	"	"
C3	1CET	"	9	2,3,4,5,6,7,11,16,17,18,19,20, 23,24,28,29,30,31,32,33,34,40	"	"	"	"
C4	1P3	"	7	2,3,4,5,6,9,11,16,17,18,19,20, 23,24,28,29,30,31,32,33,34,40	"	"	"	"
C5	1P2	"	6	2,3,4,5,7,9,11,16,17,18,19,20, 23,24,28,29,30,31,32,33,34,40	"	"	"	"
C6	1P1	"	5	2,3,4,6,7,9,11,16,17,18,19,20, 23,24,28,29,30,31,32,33,34,40	"	"	"	"
C7	1PO	"	4	2,3,5,6,7,9,11,16,17,18,19,20, 23,24,28,29,30,31,32,33,34,40	"	"	"	"
C8	1CP	"	3	2,4,5,6,7,9,11,16,17,18,19,20, 23,24,28,29,30,31,32,33,34,40	"	"	"	"
C9	1MR	"	2	3,4,5,6,7,9,11,16,17,18,19,20, 23,24,28,29,30,31,32,33,34,40	"	"	"	"
C10	2PE	"	24	2,3,4,5,6,7,9,11,16,17,18,19,20, 23,28,29,30,31,32,33,34,40	"	2	"	"
C11	2CEP	"	34	2,3,4,5,6,7,9,11,16,17,18,19,20, 23,24,28,29,30,31,32,33,40	"	2	"	"
C12	2CETA	"	33	2,3,4,5,6,7,9,11,16,17,18,19,20, 23,24,28,29,30,31,32,34,40	"	1	"	"
C13	2P7	"	40	2,3,4,5,6,7,9,11,16,17,18,19,20, 23,24,28,29,30,31,32,33,34	"	"	"	"
C14	2P6	"	20	2,3,4,5,6,7,9,11,16,17,18,19, 23,24,28,29,30,31,32,33,34,40	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C15	2P5 Input breakdown current	19	2,3,4,5,6,7,9,11,16,17,18,20,23,24,28,29,30,31,32,33,34,40	0	1	mA	7
C16	2P4 "	18	2,3,4,5,6,7,9,11,16,17,19,20,23,24,28,29,30,31,32,33,34,40	"	"	"	"
C17	2CETB "	23	2,3,4,5,6,7,9,11,16,17,18,19,20,24,28,29,30,31,32,33,34,40	"	"	"	"
C18	2P3 "	28	2,3,4,5,6,7,9,11,16,17,18,19,20,23,24,29,30,31,32,33,34,40	"	"	"	"
C19	2P2 "	29	2,3,4,5,6,7,9,11,16,17,18,19,20,23,24,28,30,31,32,33,34,40	"	"	"	"
C20	2P1 "	30	2,3,4,5,6,7,9,11,16,17,18,19,20,23,24,28,29,31,32,33,34,40	"	"	"	"
C21	2P0 "	31	2,3,4,5,6,7,9,11,16,17,18,19,20,23,24,28,29,30,32,33,34,40	"	"	"	"
C22	2CP "	17	2,3,4,5,6,7,9,11,16,18,19,20,23,24,28,29,30,31,32,33,34,40	"	2	"	"
C23	2MR "	16	2,3,4,5,6,7,9,11,17,18,19,20,23,24,28,29,30,31,32,33,34,40	"	2	"	"

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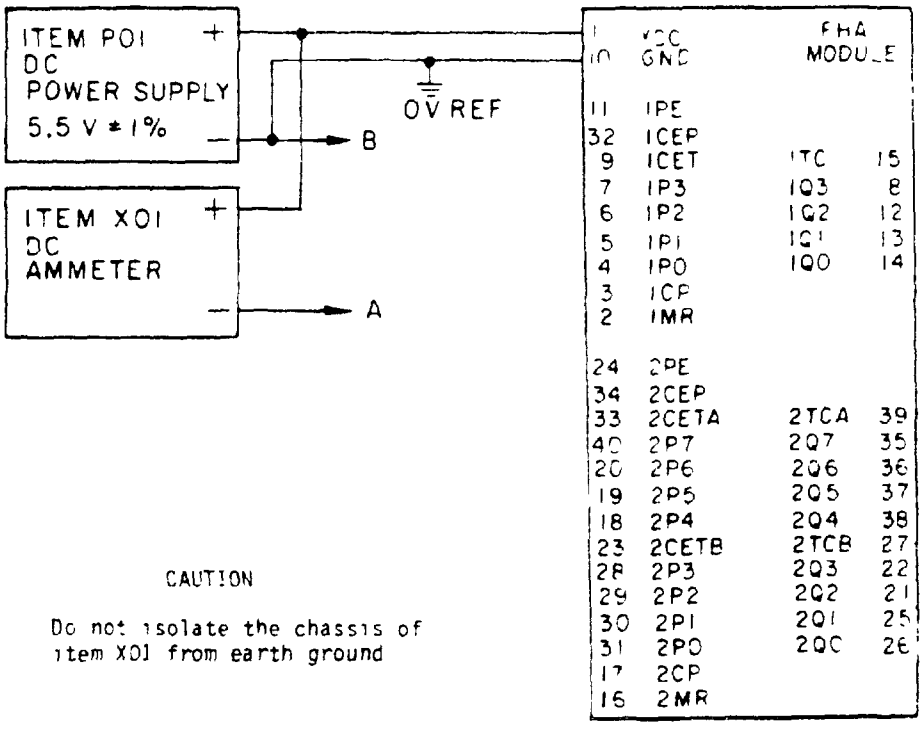


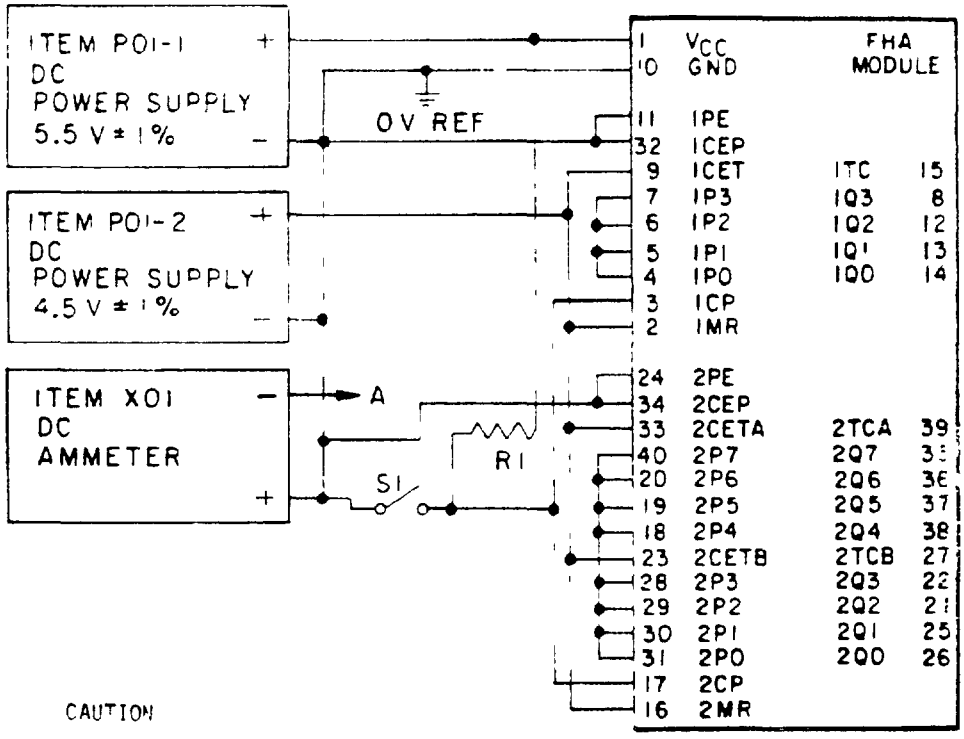
FIGURE 7 Input breakdown current test circuit for table I

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TABLE VI. Output short circuit current test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D1	1TC	Output short circuit current	15	-40	-100	mA	8
D2	1Q3	"	8	"	"	"	"
D3	1Q2	"	12	"	"	"	"
D4	1Q1	"	13	"	"	"	"
D5	1Q0	"	14	"	"	"	"
D6	2TCA	"	39	"	"	"	"
D7	2Q7	"	35	"	"	"	"
D8	2Q6	"	36	"	"	"	"
D9	2Q5	"	37	"	"	"	"
D10	2Q4	"	38	"	"	"	"
D11	2TCB	"	27	"	"	"	"
D12	2Q3	"	22	"	"	"	"
D13	2Q2	"	21	"	"	"	"
D14	2Q1	"	25	"	"	"	"
D15	2Q0	"	26	"	"	"	"

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CAUTION

Do not isolate the chassis of item X01 from earth ground

1. R1 = 1 kΩ ±5% 1/4 W.
2. S1 = SPST switch.
3. When performing output short circuit current test, only one output shall be shorted at a time and the duration of the test shall not exceed 5 seconds.
4. The following sequence must be completed before performing test conditions D1-D15

   a. With S1 open and tied to pins 3 and 17, connect item P01-2 to pins 4, 5, 6, 7, 31, 30, 29, 28, 18, 19, 20, and 40.
   b. Close S1 and open
   c. Close S1 and connect pins 4, 5, 6, 7, 31, 30, 29, 28, 18, 19, 20, and 40 to 0 V REF and perform test conditions D1-D15 per table VI.

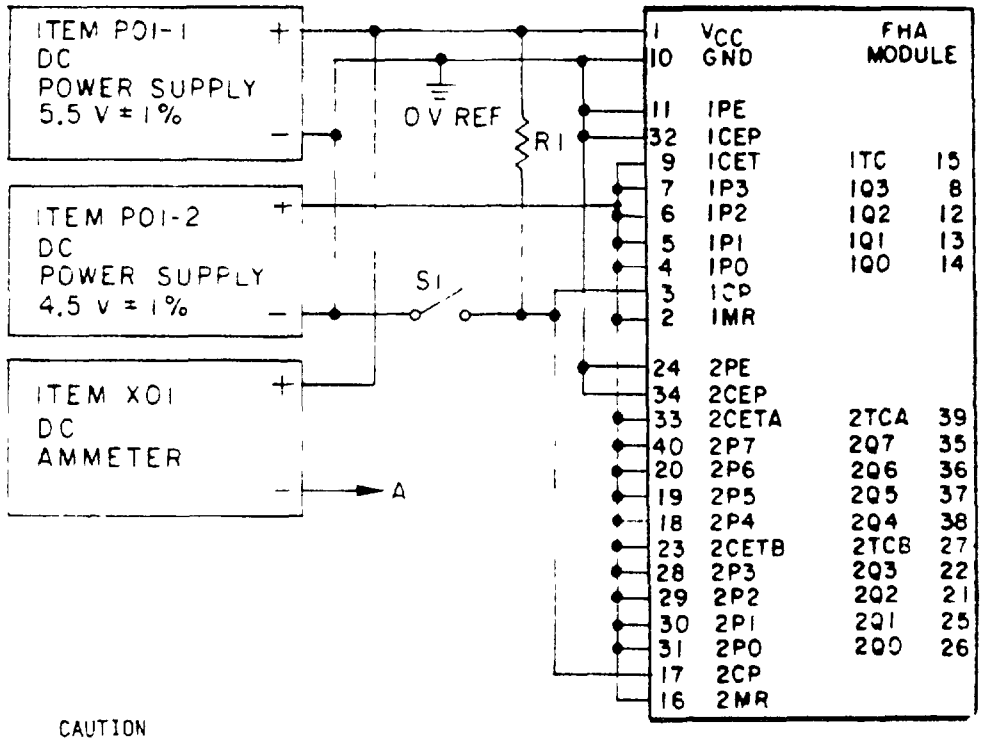
FIGURE 8 Output short circuit current test circuit for table VI

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TABLE VII. Output leakage current test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
E1	1TC	Output leakage current	15	0	1	mA	9
E2	1Q3	"	8	"	"	"	"
E3	1Q2	"	12	"	"	"	"
E4	1Q1	"	13	"	"	"	"
E5	1Q0	"	14	"	"	"	"
E6	2TCA	"	39	"	"	"	"
E7	2Q7	"	35	"	"	"	"
E8	2Q6	"	36	"	"	"	"
E9	2Q5	"	37	"	"	"	"
E10	2Q4	"	38	"	"	"	"
E11	2TCB	"	27	"	"	"	"
E12	2Q3	"	22	"	"	"	"
E13	2Q2	"	21	"	"	"	"
E14	2Q1	"	25	"	"	"	"
E15	2Q0	"	26	"	"	"	"

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CAUTION

Do not isolate the chassis of item X01 from earth ground

NOTES:
1. R1 = [illegible] ±5 , 1/4 W.
2. S1 = SPST switch
3. Before performing test conditions E1-E15 switch S1 must be closed then opened.

FIGURE 9 Output leakage current test circuit for table VII

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TABLE VIII. Input clamp diode voltage test.

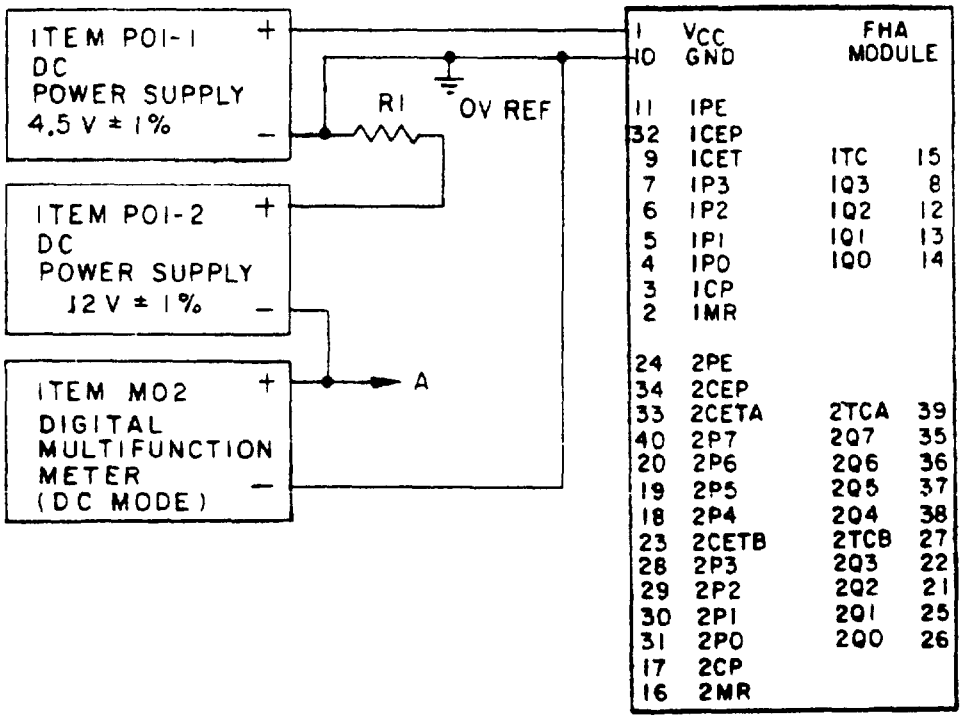
Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
F1	1PE	Input clamp diode voltage	11	-0.4	-1.2	V	10
F2	1CEP	"	32	"	"	"	"
F3	1CET	"	9	"	"	"	"
F4	1P3	"	7	"	"	"	"
F5	1P2	"	6	"	"	"	"
F6	1P1	"	5	"	"	"	"
F7	1P0	"	4	"	"	"	"
F8	1CP	"	3	"	"	"	"
F9	1MR	"	2	"	"	"	"
F10	2PE	"	24	"	"	"	"
F11	2CEP	"	34	"	"	"	"
F12	2CETA	"	33	"	"	"	"
F13	2P7	"	40	"	"	"	"
F14	2P6	"	20	"	"	"	"
F15	2P5	"	19	"	"	"	"
F16	2P4	"	18	"	"	"	"
F17	2CETB	"	23	"	"	"	"
F18	2P3	"	28	"	"	"	"
F19	2P2	"	29	"	"	"	"
F20	2P1	"	30	"	"	"	"

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TABLE VIII. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
F21	2PO	Input clamp diode voltage	31	-0.4	-1.2	V	10
F22	2CP	"	17	"	"	"	"
F23	2MR	"	16	"	"	"	"

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NOTE R1 = 590Ω ±1%, 1/4 W

* FIGURE 10 Input clamp diode voltage test circuit for table VIII

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TABLE IX Functional test 1/-3/

Test condition	Parameter	Connect			Output logic states															Test figure
		A to pin	B to pin	D to pin	1Q0	1Q1	1Q2	1Q3	1TC	2Q4	2Q5	2Q6	2Q7	2TCA	2Q0	2Q1	2Q2	2Q3	2TCB	
G1	Functional	2,32,9,16,34,33 23	4,5,6 7,11 18,19, 20,40,24,31,30,29, 28	3,17	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	11 12
G2	"	2,11,32,9,16,24, 34,33,23	4,5,6,7,18,19,20, 40 31,30 29,28	"	H	L	L	L	L	H	L	L	L	L	H	L	L	L	L	"
G3	"	"	"	"	L	H	L	L	L	L	H	L	L	L	L	H	L	L	L	"
G4	"	"		"	H	H	L	L	L	H	H	L	L	L	H	H	L	L	L	"
G5	"	"	"	"	L	L	H	L	L	L	L	H	L	L	L	L	H	H	L	"
G6	"	"		"	H	L	H	L	L	H	L	H	L	L	H	L	H	L	L	"
G7	"	"	"	"	L	H	H	L	L	L	H	H	L	L	L	H	H	H	L	"
G8	"	"	"	"	H	H	H	L	L	H	H	H	L	L	H	H	H	L	L	"
G9	"	"	"	"	L	L	L	H	L	L	L	L	H	L	L	L	L	H	L	"
G10	"	"	"	"	H	L	L	H	L	H	L	L	H	L	H	L	L	H	L	"
G11	"	"	"	"	L	H	L	H	L	L	H	L	H	L	L	H	L	H	L	"
G12	"	"	"	"	H	H	L	H	L	H	H	L	H	L	H	H	L	H	L	"
G13	"	"			L	L	H	H	L	L	L	H	H	L	L	L	H	H	L	"
G14	"	"		"	H	L	H	H	L	H	L	H	H	L	H	L	H	H	L	"
G15	"	"		"	L	H	H	H	L	L	H	H	H	L	L	H	H	H	L	"
G16	"	"	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
G17	"	"		"	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
G18	"	4,5,6,7,11,32,18, 19,20,40,24,34,31, 30,29,28	2,9,16,33,23	"	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
G19	"	4,5,6,7,2,11,32, 18,19,20,40,16,24 34,31,30,29,28	9 33 23	"	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	

See footnotes at end of table

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TABLE IX. Functional test 1/ 2/ Continued

Test condition	Parameter	Connect			Output logic states															Test figure
		A to pin	B to pin	D to pin	I00	I01	I02	I03	I1C	I04	I05	I06	I07	I2TCA	I00	I01	I02	I03	I2TCB	
G20	Functional	4,5,6,7,32,18,19, 20,40,34,31,30,29, 28	2 11,9,16,24,33,23	3,17	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	11,12
G21	"	4,5,6,7,2,32,18, 19,20,40,16,34,31, 30,29,28	11,9,24, 3,23		H	H	H	H	L	H	H	H	H	L	H	H	H	H	L	"
G22	"	4,5,6,7,11,9,18, 19,20,40,24,33,31, 30,29,28 23	2,32,16,34	"	L	L	"	L	L	L	L	L	L	L	L	L	L	L	L	"
G23	"	4,5,6,7,2,11,9,18 19,20,40,16,24,33, 31,30,29,28,23	32,34	"	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
G24	"	5,2,32,9,19,16,34, 33,30,23	4,6,7,11,18,20,40, 24,31 29 28	"	L	H	L	L	L	L	H	L	L	L	L	H	L	L	L	"
G25	"	2,32,9,16,34, 33,23	4,5,6,7,11,18,19, 20 40 24 31,30,29, 28	"	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
G26	"	6,2,32,9,20,16,34, 33,29,23	4,5,7,11,18,19,40, 24,31 30,28	"	L	L	H	L	L	L	L	H	L	L	L	L	H	L	L	"
G27	"	2,32,9,16,34,33,23	4,5,6 7,11,18,19, 20,40,24,31,30,29, 28,	"	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
G28	"	7,2,32,9,40,16, 34,33,28,23	4,5,6,11,18,19, 20,24 31,30,29	"	L	L	L	H	L	L	L	L	H	L	L	L	L	H	L	"
G29	"	2,32,9,16,34,33,23	4,5,6 7,11,18,19, 20,40 24,31,30,29, 28	"	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
G30	"	4,2,9,18,16,33,31, 23	5,6,7,11,32,19,20, 40,24,34 30,29,28,	"	H	L	L	L	L	H	L	L	L	L	H	L	L	L	L	"
G31	"	4,2,11,9,18,16,24, 33,31,23	5,6,7,32,19,20,40, 34,30 29,28,	"	H	L	L	L	L	H	L	L	L	L	H	L	L	L	L	"

See footnotes at end of table.

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TABLE IX. Functional test 1/-3/ - Continued

Test condition	Parameter	Connect			Output logic states														Test figure
		A to pin	B to pin	D to pin	1Q0	1Q1	1Q2	1Q3	1TC	2Q4	2Q5	2Q6	2Q7	2TCA	2Q0	2Q1	2Q2	2Q3	2TCB	
G32	Functional	2,16	4,5,6,7,11,32,9, 18,19,20,40,24,34 33,31,30,29,28,23	3,17	L	L	L	H	L	L	L	L	L	L	L	L	L	L	L	11,12
G33	"	1,5,2,18,19,16, 31,30	6,7,11,32,9,20,40 24,34,33,29,28,23	"	H	H	L	L	L	H	H	L	L	L	H	H	L	L	L	"
G34		2,16	4,5,6,7,11,32,9, 18,19,20,40,24,34 33,31,30,29,28,23		L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
G35	"	4,5,6,2,18,19,20 16,31,30,29	7,11,32,9,40,24 34,33,28,24,23	"	H	H	H	L	L	H	H	H	L	L	H	H	H	L	L	"
G36	"	2,16	4,5,6,7,11,32,9 18,19,20,40,24,34 33,31,30,29,28,23		L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
G37	"	1,5,6,7,2,18,19, 20,40,16,31,30,29, 28	11,32,9,24,34 33,23	"	L	H	H	H	L	H	H	H	H	L	H	H	H	H	L	"
38		4,5,6,7,2,11,18, 19,20,40,16,24,31, 30,29,28	32,9,34,33,23	"	H	H	H	H	L	H	H	H	H	L	H	H	H	H	L	"
G39		4,5,6,7,2,11,18, 19,20,40,16,24,31, 30,29,28	32,9,34,33,23		H	H	H	H	L	H	H	H	H	L	H	H	H	H	L	"

1/ Output logic states
L - Low level 0.4 V
H - High level 2.4 V

2/ For test conditions G1 G39 verification of the output logic states in footnote 1/ is obtained by connecting item M02, point C, to each output in succession after input conditions have been accomplished. Set item 101 to the single pulse mode and pulse once before each set of readings. Test conditions G1 G39 are to be performed in sequence.

3/ 1Q0, 1Q1, 1Q2, 1Q3, 1TC, 2Q4, 2Q5, 2Q6, 2Q7, 2TCA, 2Q0, 2Q1, 2Q2, 2Q3 and 2TCB output logic states are measured at output pins 14, 13, 12, 8, 15, 38, 37, 36, 35, 19, 26, 25, 21, 2, and 27 respectively.

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*TABLE X. Output voltage (low-level and high-level) test.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H1	1TC	Output voltage, low level	4,5,6,7,32	9,11,E	15	3	B	2	0	500	mV	11,12
H2	1Q3	"	4,5,6,32,9	E,7,11	8	"	"	"	"	"	"	"
H3	1Q2	"	4,5,7,32,9	E,6,11	12	"	"	"	"	"	"	"
H4	1Q1	"	4,6,7,32,9	E,5,11	13	"	"	"	"	"	"	"
H5	1Q0	"	5,6,7,32,9	E,4,11	14	"	"	"	"	"	"	"
H6	2TCA	"	18,19,20, 40,34	24,33,E	39	17	"	16	"	"	"	"
H7	2Q7	"	18,19,20, 33,34	24,40,E	35	"	"	"	"	"	"	"
H8	2Q6	"	18,19,40, 33,34	24,20,E	36	"	"	"	"	"	"	"
H9	2Q5	"	18,20,40, 33,34	24,19,E	37	"	"	"	"	"	"	"
H10	2Q4	"	19,20,40, 33,34	24,18,E	38	"	"	"	"	"	"	"
H11	2TCB	"	28,29,30, 31,34	24,23,E	27	"	"	"	"	"	"	"
H12	2Q3	"	29,30,34, 23,31	24,28,E	22	"	"	"	"	"	"	"
H13	2Q2	"	28,30,34, 31,23	24,29,E	21	"	"	"	"	"	"	"
H14	2Q1	"	28,29,31, 34,23	24,30,E	25	"	"	"	"	"	"	"
H15	2Q0	"	28,29,30, 34,23	24,31,E	26	"	"	"	"	"	"	"

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*TABLE X. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
J1	1TC	Output voltage, high level	4,5,6,7, 9,32	11,E	15	3	8	2	2.5	4.5	V	11,12
J2	1Q3	"	7	4,5 6,9, 32,11,E	8	"	"	"	"	"	"	"
J3	1Q2	"	6	4,5,7,9, 32,11,E	12	"	"	"	"	"	"	"
J4	1Q1	"	5	4,6,7,9, 32,11,E	13	"	"	"	"	"	"	"
J5	1Q0	"	4	5,6,7,9, 32,11,E	14	"	"	"	"	"	"	"
J6	2TCA	"	18,19,20, 40,33,34	24,E	39	17	"	16	"	"	"	"
J7	2Q7	"	40	18,19,20, 33,34,24,F	35	"	"	"	"	"	"	"
J8	2Q6	"	20	18,19,40, 33,34,24,E	36	"	"	"	"	"	"	"
J9	2Q5	"	19	18,20,40, 33,34,24,E	37	"	"	"	"	"	"	"
J10	2Q4	"	18	19,20,40, 33,34,24,E	38	"	"	"	"	"	"	"
J11	2TCB	"	28,29,30, 31,23,34	24,E	27	"	"	"	"	"	"	"
J12	2Q3	"	28	29,30,31, 23,24,34,E	22	"	"	"	"	"	"	"
J13	2Q2	"	29	28,30,31, 23,24,34,F	21	"	"	"	"	"	"	"

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*TABLE X Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
J14	2Q1 Output voltage, high level	30	28,29,31, 23,24,34,E	25	17	B	16	2.5	4 5	V	11,12
J15	2Q0 Output voltage, high level	31	28,29,30, 23,24,34,E	26	17	B	16	2.5	4.5	V	11,12

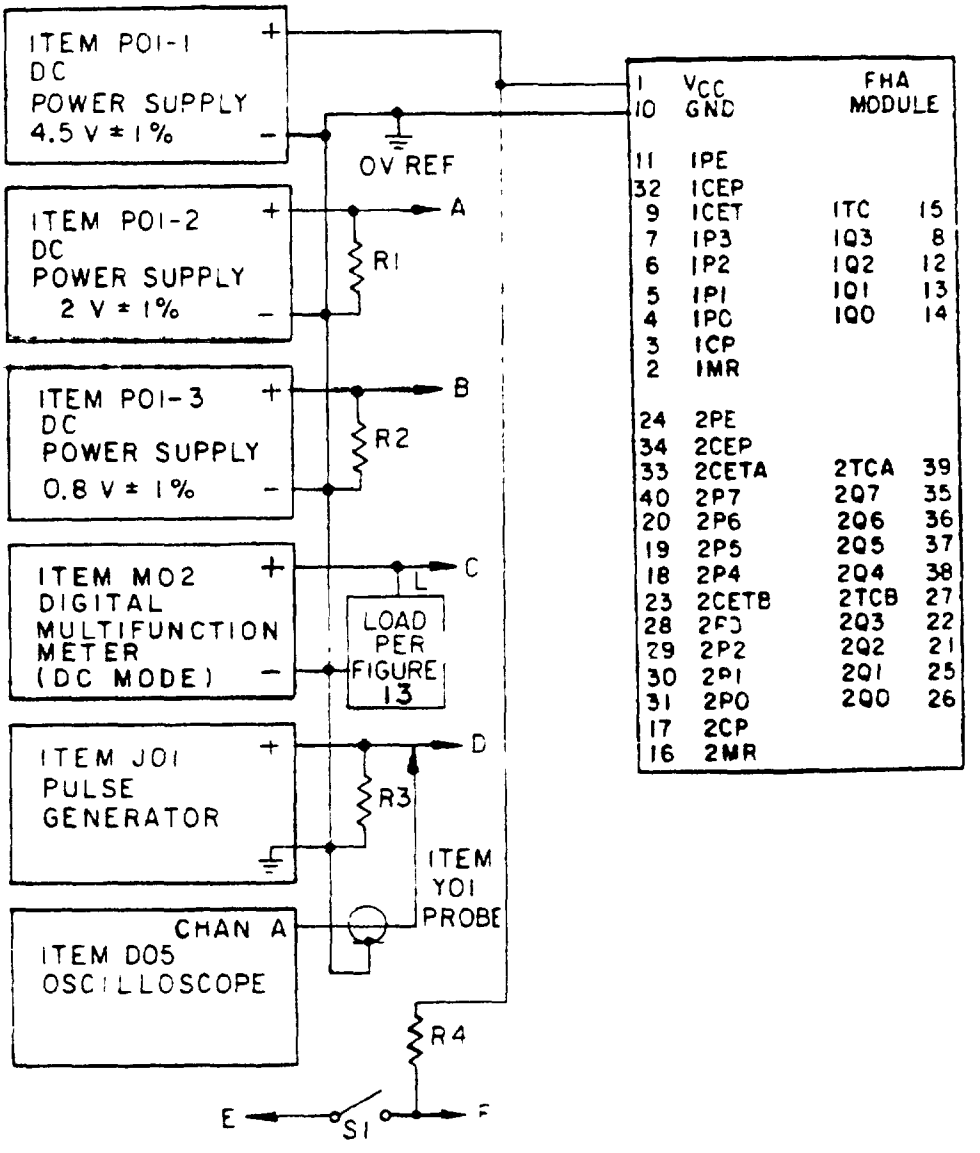
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*TABLE XI Input threshold (low-level and high level) test. 1/

Test condition	Parameter		Connect						Output logic states					Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	1TC	1Q3	1Q2	1Q1	1Q0	
K1	1MR	Input threshold, low level	4,5,6,7, 9,32	11,E	Open	2	B	3	L	L	L	L	L	11,12
K2	1CET, 1CEP, 1PE	Input threshold, high level	9,11,32	,F,4,5,6	"	3	"	2	L	L	L	L	H	"
									2TCA	2Q7	2Q6	2Q5	2Q4	
K3	2MR	Input threshold, low level	18,19,20, 40,33,34	24,E	"	16	"	17	L	L	L	L	L	"
K4	2CETA, 2CEP, 2PE	Input threshold, high level	33,24,34	18,19,20, 40,E	"	17	"	16	L	L	L	L	H	"
									2TCB	2Q3	2Q2	2Q1	2Q0	
K5	2MR	Input threshold, low level	23,28,29, 30,31,34	24,E	"	16	"	17	L	L	L	L	L	"
K6	2CETB, 2CEP, 2PE	Input threshold, high level	23,24,34	28 29,30, 31 E	"	17	"	16	L	L	L	L	H	"

1/ Output logic states
L = Low level = 0 to 500 mV
H = High level = 2 5 to 4.5 V.

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NOTES
1. R1 - R3 = 51Ω ±5%, 1/4 W.
2. R4 = 1 kΩ ±5%, 1/4 W.
3. S1 = SPST switch.
4. Test conditions H1-H15, J1-J15, K2, K4, and K6.

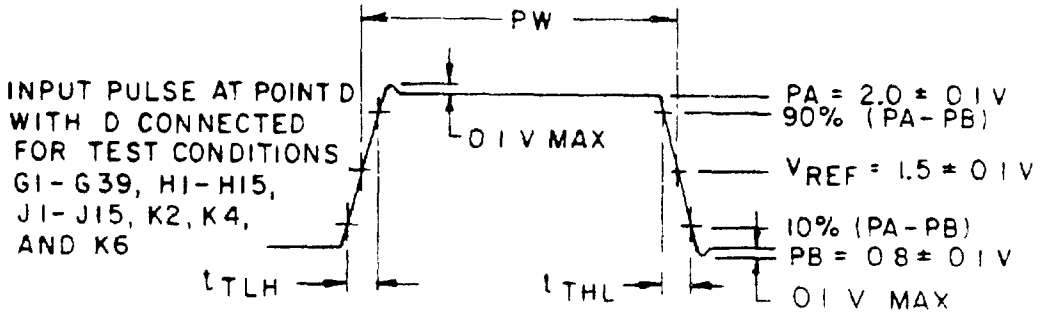
After making all circuit connections for a test condition, close switch S1 for a minimum of 100 ns then open switch S1. With item J01 in the single pulse mode, apply one pulse to the circuit under test. The voltage levels and output logic states shall be as specified in tables X and XI.

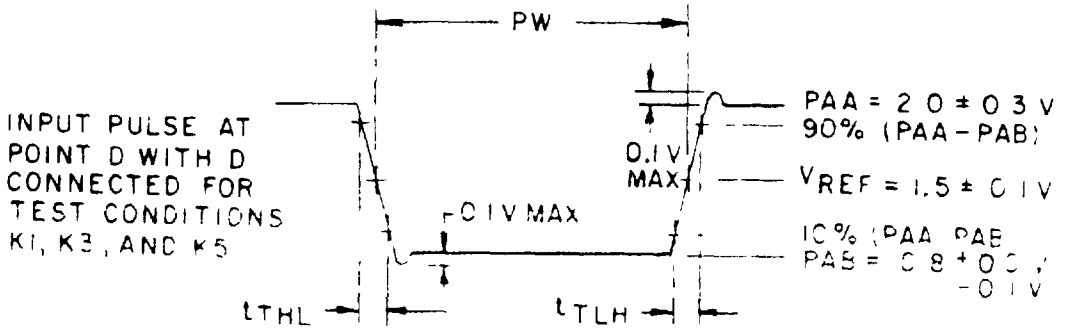
5. Test conditions K1, K3, and K5

Initially switch S1 shall be closed. After making all circuit connections for a test condition, open switch S1 for a minimum of 100 ns then close switch S1. With item J01 in the single pulse mode, apply one pulse to the circuit under test. The output logic states shall be as specified in table XI.

* FIGURE 11. Functional, output voltage (low-level and high-level), and input threshold (low-level and high-level) test circuit for tables IX, X, and XI.

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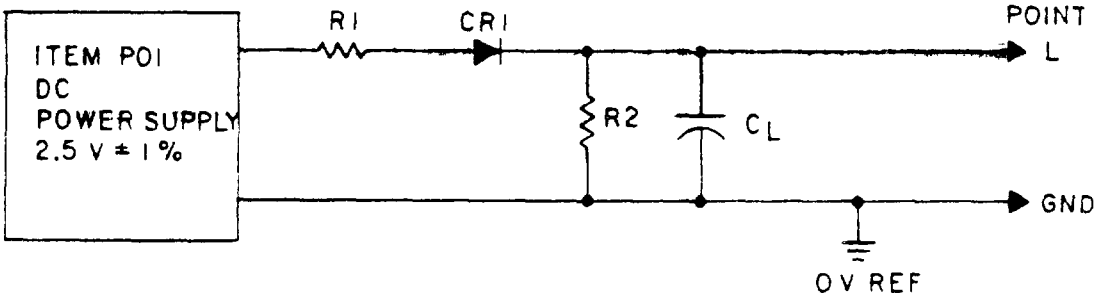


NOTE

Input pulse parameters
PW = 500 ±100 ns.
$t_{TLH} = t_{THL}$ = 30 ±5 ns.

* FIGURE 12 Functional, output voltage (low-level and high-level), and input threshold (low-level and high-level) test circuit waveforms for tables IX, X, and XI

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NOTES.
1. CR1 = 1N3600, 1N4150, or equivalent
2 R1 = Select value for a current flow of 20 mA ±5 out of point L with point L held at 0.5 V.
3 R2 = 2.49 k ±1%, 1/8 W
4 Test type S

C_L = 50 ±5 pF (including probe and parasitic capacitance).

Test type O

C_L = 150 ±5 pF (including probe and parasitic capacitance)

5 Capacitor C_L is not required for dc electrical test conditions G1-G39, H1-H15, J1-J15, K1-K6, and L1-L22
6 Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm)

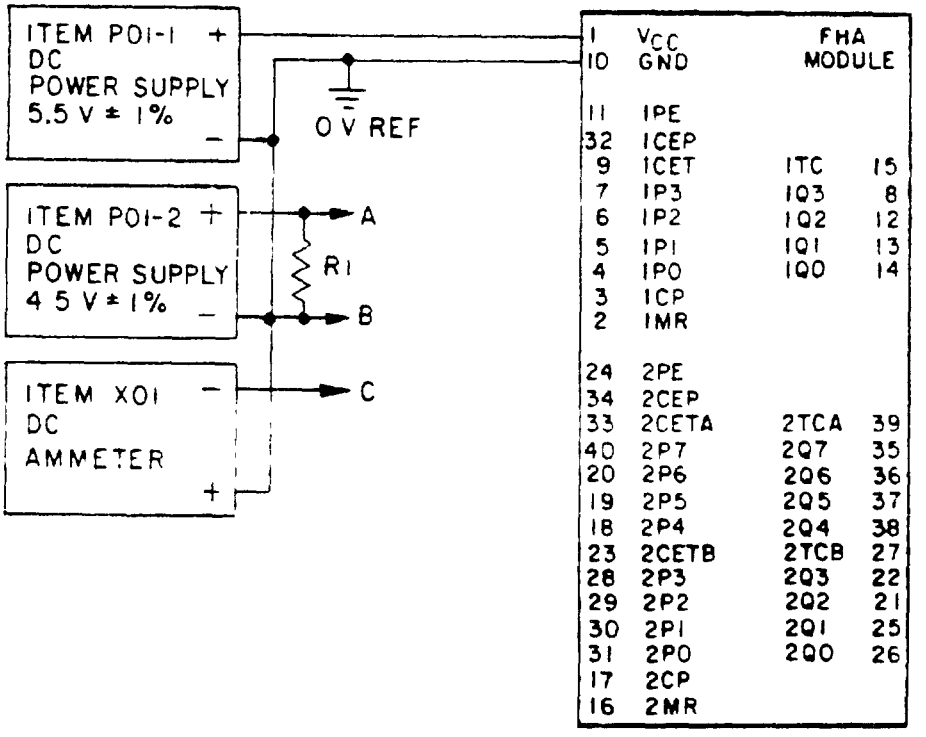
* FIGURE 13. FHA logic load circuit.

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TABLE XII Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	1PE Input current, low level	9,32	Open	11	-0.1	-4.4	mA	14
L2	1CEP "	9,11	"	32	"	"	"	"
L3	1CET "	11,32	"	9	"	"	"	"
L4	1P3 "	Open	11	7	"	-2.2	"	"
L5	1P2 "	"	"	6	"	"	"	"
L6	1P1 "	"	"	5	"	"	"	"
L7	1P0 "	"	"	4	"	"	"	"
L8	1CP "	"	Open	3	"	-5.5	"	"
L9	1MR "	"	"	2	"	-2.2	"	"
L10	2PE "	23,33,34	"	24	"	-8.8	"	"
L11	2CEP "	23,24,33	"	34	"	-8.8	"	"
L12	2CETA "	24,34	"	33	"	-4.4	"	"
L13	2P7 "	Open	24	40	"	-2.2	"	"
L14	2P6 "	"	"	20	"	"	"	"
L15	2P5 "	"	"	19	"	"	"	"
L16	2P4 "	"	"	18	"	"	"	"
L17	2CETB "	24,34	Open	23	"	-4.4	"	"
L18	2P3 "	Open	24	28	"	-2.2	"	"
L19	2P2 "	"	"	29	"	"	"	"
L20	2P1 "	"	"	30	"	"	"	"
L21	2P0 "	"	"	31	"	"	"	"
L22	2CP "	"	Open	17	"	-11.0	"	"
L23	2MR "	"	Open	16	"	-5.1	"	"

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CAUTION

Do not isolate the chassis of item X01 from earth ground

NOTE R1 = 51Ω ±5%, 1/2 W.

FIGURE 14 Input current (low-level) test circuit for table XII.

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TABLE VIII. Input current (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	1PE Input current, high level	Open	2,3,4,5,6,7,9,16,17,18, 19,20,23,24,28,29,30,31, 32,33,34,40	11	-1	100	µA	15
M2	1CEP "	"	2,3,4,5,6,7,9,11,16,17, 18,19,20,23,24,28,29,30, 31,33,34,40	32	"	"	"	"
M3	1CET "	"	2,3,4,5,6,7,11,16,17,18, 19,20,23,24,28,29,30,31, 32,33,34,40	9	"	"	"	"
M4	1P3 "	11	2,3,4,5,6,9,16,17,18, 19,20,23,24,28,29,30,31, 32,33,34,40	7	"	50	"	"
M5	1P2 "	"	2,3,4,5,7,9,16,17,18,19, 20,23,24,28,29,30,31,32, 33,34,40	6	"	"	"	"
M6	1P1 "	"	2,3,4,6,7,9,16,17,18,19, 20,23,24,28,29,30,31,32, 33,34,40	5	"	"	"	"
M7	1P0 "	"	2,3,5,6,7,9,16,17,18,19, 20,23,24,28,29,30,31,32, 33,34,40	4	"	"	"	"
M8	1CP "	Open	2,4,5,6,7,9,11,16,17,18, 19,20,23,24,28,29,30,31, 32,33,34,40	3	"	125	"	"
M9	1MR "	"	3,4,5,6,7,9,11,16,17,18, 19,20,23,24,28,29,30,31, 32,33,34,40	2	"	50	"	"
M10	2PE "	"	2,3,4,5,6,7,9,11,16,17, 18,19,20,23,28,29,30,31, 32,33,34,40	24	"	200	"	"

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TABLE XIII Input current (high-level) test - Continued

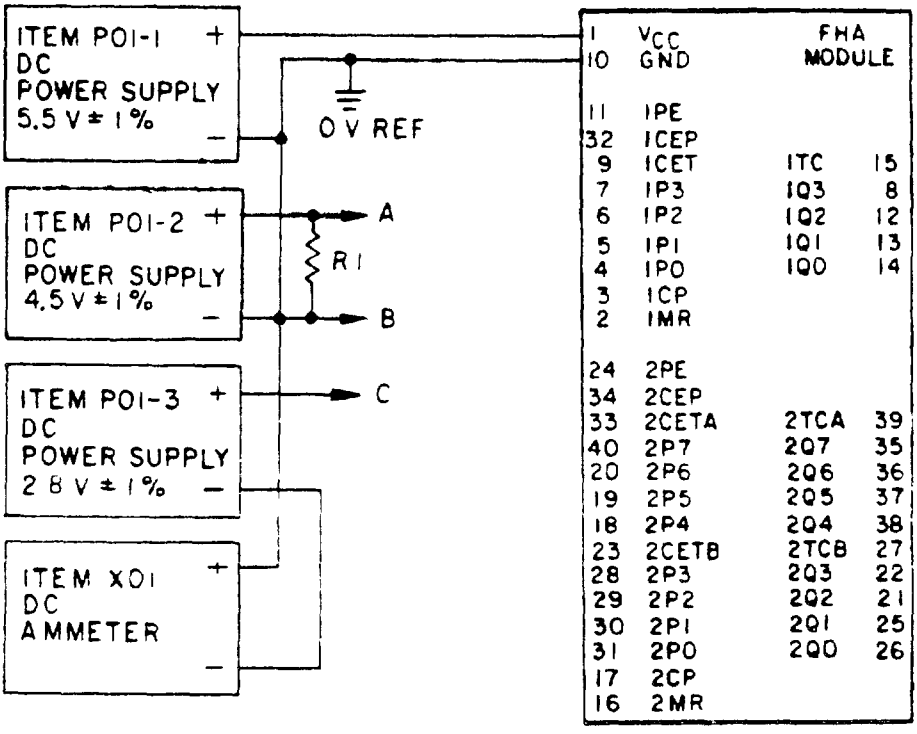
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M11	2CEP Input current, high level	Open	2,3,4,5,6,7,9,11,16,17, 18,19,20,23,24,28,29,30, 31,32,33,40	34	-1	200	uA	15
M12	2CETA "	Open	2,3,4,5,6,7,9,11,16,17, 18,19,20,23,24,28,29,30, 31,32,34,40	33	"	100	"	"
M13	2P7 "	24	2,3,4,5,6,7,9,11,16,17, 18,19,20,23,28,29,30, 31,32,33,34	40	"	50	"	"
M14	2P6 "	"	2,3,4,5,6,7,9,11,16,17, 18,19,23,28,29,30,31, 32,33,34,40	20	"	"	"	"
M15	2P5 "	"	2,3,4,5,6,7,9,11,16,17, 18,20,23,28,29,30,31, 32,33,34,40	19	"	"	"	"
M16	2P4 "	"	2,3,4,5,6,7,9,11,16,17, 19,20,23,28,29,30,31, 32,33,34,40	18	"	"	"	"
M17	2CETB "	Open	2,3,4,5,6,7,9,11,16,17, 18,19,20,24,28,29,30,31, 32,33,34,40	23	"	100	"	"
M18	2P3 "	24	2,3,4,5,6,7,9,11,16,17, 18,19,20,23,29,30,31, 32,33,34,40	28	"	50	"	"
M19	2P2 "	"	2,3,4,5,6,7,9,11,16,17, 18,19,20,23,28,30,31, 32,33,34,40	29	"	"	"	"
M20	2P1 "	"	2,3,4,5,6,7,9,11,16,17, 18,19,20,23,28,29,31, 32,33,34,40	30	"	"	"	"
M21	2P0 "	"	2,3,4,5,6,7,9,11,16,17, 18,19,20,23,28,29,30, 32,33,34,40	31	"	"	"	"

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TABLE XIII. Input current (high-level) test - Continued

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M22	2CP Input current, high level	Open	2,3,4,5,6,7,9,11,16,18, 19,20,23,24,28,29,30, 31,32,33,34,40	17	-1	250	µA	15
M23	2MR Input current, high level	Open	2,3,4,5,6,7,9,11,17,18, 19,20,23,24,28,29,30, 31,32,33,34,40	16	-1	-315	µA	15

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CAUTION

Do not isolate the chassis of item X01 from earth ground

NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 15. Input current (high-level) test circuit for table XIII.

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TABLE XIV. Power supply current (low-level outputs) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N	Power supply current, low level outputs	2,16	3,4,5,6,7,9,11,17, 18,19,20,23,24,28, 29,30,31,32,33,34,40	50	382	mA	16

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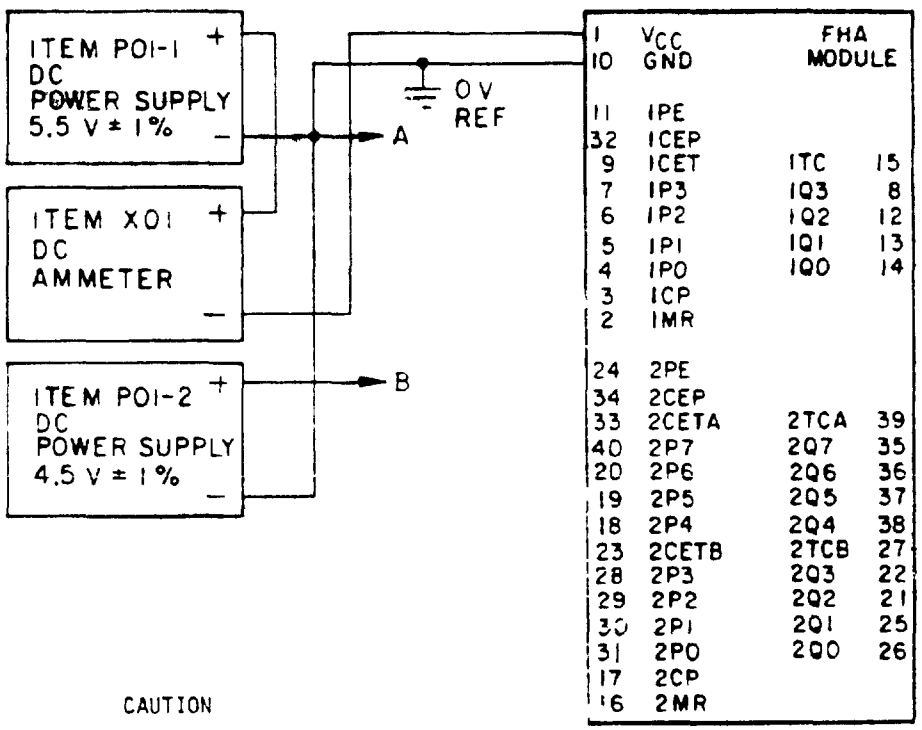


FIGURE 16 Power supply current (low-level outputs) test circuit for table XIV.

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TABLE XV Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
P1	1TC Transition time (t_{TLH}), C_L = 50 pF	15	4	17	ns	17,18
P2	1Q3 "	8	"	"	"	"
P3	1Q2 "	12	"	"	"	"
P4	1Q1 "	13	"	"	"	"
P5	1Q0 "	14	"	"	"	"
P6	2TCA "	39	"	"	"	"
P7	2Q7 "	35	"	"	"	"
P8	2Q6 "	36	"	"	"	"
P9	2Q5 "	37	"	"	"	"
P10	2Q4 "	38	"	"	"	"
P11	2TCB "	27	"	"	"	"
P12	2Q3 "	22	"	"	"	"
P13	2Q2 "	21	"	"	"	"
P14	2Q1 "	25	"	"	"	"
P15	2Q0 "	26	"	"	"	"
P16	1TC Transition time (t_{THL}), C_L = 150 pF	15	6	27	"	"
P17	1Q3 "	8	"	"	"	"
P18	1Q2 "	12	"	"	"	"
P19	1Q1 "	13	"	"	"	"
P20	1Q0 "	14	"	"	"	"
P21	2TCA "	39	"	"	"	"
P22	2Q7 "	35	"	"	"	"

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TABLE XV Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
P23	2Q6 Transition time (t_{TLH}), C_L = 150 pF	36	6	27	ns	17,18
P24	2Q5 "	37	"	"	"	"
P25	2Q4 "	38	"	"	"	"
P26	2TCB "	27	"	"	"	"
P27	2Q3 "	22	"	"	"	"
P28	2Q2 "	21	"	"	"	"
P29	2Q1 "	25	"	"	"	"
P30	2Q0 "	26	"	"	"	"
Q1	1TC Transition time (t_{THL}) (C_L = 50 pF	15	2	9	"	"
Q2	1Q3 "	8	"	"	"	"
Q3	1Q2 "	12	"	"	"	"
Q4	1Q1 "	13	"	"	"	"
Q5	1Q0 "	14	"	"	"	"
Q6	2TCA "	39	"	"	"	"
Q7	2Q7 "	35		"	"	"
Q8	2Q6 "	36	"	"	"	"
Q9	2Q5 "	37	"	"	"	"
Q10	2Q4 "	38	"	"	"	"
Q11	2TCB "	27	"	"	"	"
Q12	2Q3 "	22	"	"	"	"

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TABLE XV Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued

Test condition	Parameter	Connect A to pin	Initial Limits 25°C ±5°C ambient Min	Initial Limits 25°C ±5°C ambient Max	Unit	Test figure
Q13	2Q2 Transition time (t_{TLH}), C_L - 50 pF	21	2	9	ns	17,18
Q14	2Q1 "	25	"	"	"	"
Q15	2Q0 "	26	"	"	"	"
Q16	1TC Transition time (t_{THL}) C_L - 150 pF	15	"	11	"	"
Q17	1Q3 "	8	"	"	"	"
Q18	1Q2 "	12	"	"	"	"
Q19	1Q1 "	13	"	"	"	"
Q20	1Q0 "	14	"	"	"	"
Q21	2TCA "	39	"	"	"	"
Q22	2Q7 "	35	"	"	"	"
Q23	2Q6 "	36	"	"	"	"
Q24	2Q5 "	37	"	"	"	"
Q25	2Q4 "	38	"	"	"	"
Q26	2TCB "	27	"	"	"	"
Q27	2Q3 "	22	"	"	"	"
Q28	2Q2 "	21	"	"	"	"
Q29	2Q1 "	25	"	"	"	"
Q30	2Q0 "	26	"	"	"	"
R1	CP to 1TC Propagation delay time (t_{PLH}), C_L - 50 pF	15	6	21	"	"
R2	CP to 1Q3 "	8	4	13	"	"
R3	CP to 1Q2 "	12	4	13	"	"

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TABLE XV Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
R4	CP to 1Q1	Propagation delay time (t_{PLH}), C_l = 50 pF	13	4	13	ns	17,18
R5	CP to 1Q0	"	14	4	13	"	"
R6	CP to 2TCA	"	39	6	21	"	"
R7	CP to 2Q7	"	35	4	13	"	"
R8	CP to 2Q6	"	36	"	"	"	"
R9	CP to 2Q5	"	37	"	"	"	"
R10	CP to 2Q4	"	38	"	"	"	"
R11	CP to 2TCB	"	27	6	21	"	"
R12	CP to 2Q3	"	22	4	13	"	"
R13	CP to 2Q2	"	21	"	"	"	"
R14	CP to 2Q1	"	25	"	"	"	"
R15	CP to 2Q0	"	26	"	"	"	"
R16	CP to 1TC	Propagation delay time (t_{PLH}), C_l = 150 pF	15	6	29	"	"
R17	CP to 1Q3	"	8	4	21	"	"
R18	CP to 1Q2	"	12	"	"	"	"
R19	CP to 1Q1	"	13	"	"	"	"
R20	CP to 1Q0	"	14	"	"	"	"
R21	CP to 2TCA	"	39	6	29	"	"
R22	CP to 2Q7	"	35	4	21	"	"
R23	CP to 2Q6	"	36	"	"	"	"
R24	CP to 2Q5	"	37	"	"	"	"

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TABLE XV Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued

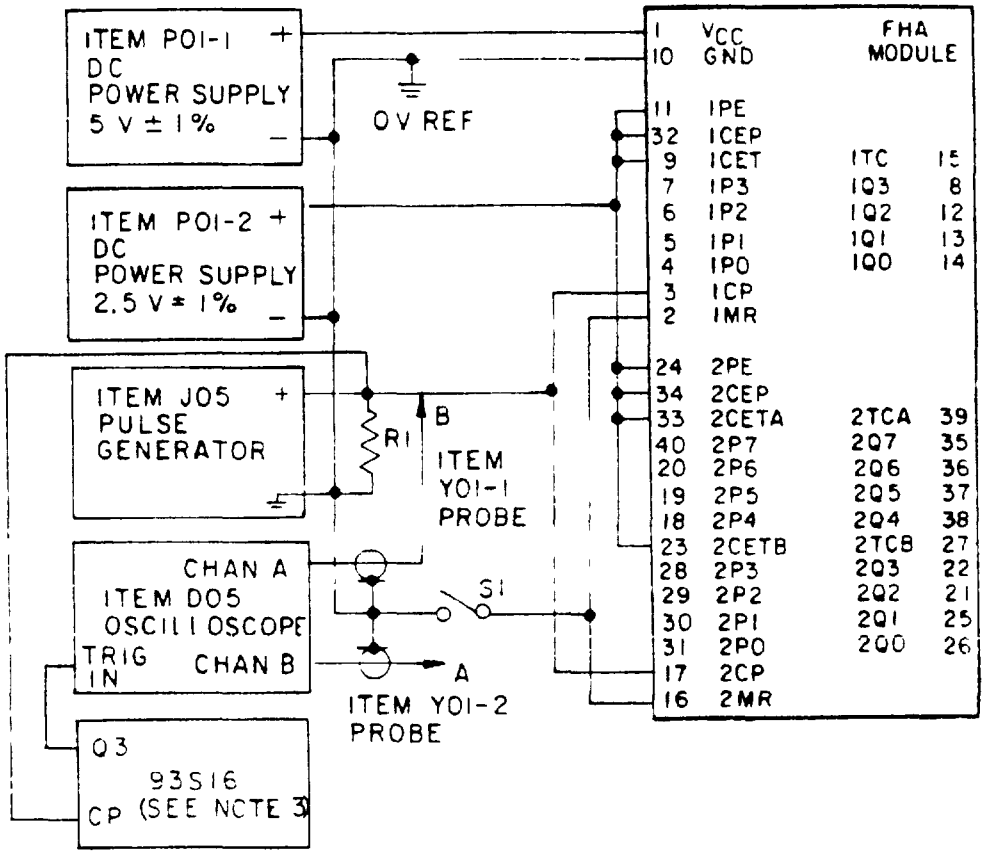
Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
R25	CP to 2Q4	Propagation delay time (t_{PLH}), C_L = 150 pF	38	4	21	ns	17,18
R26	CP to 2TCB	"	27	6	29	"	"
R27	CP to 2Q3	"	22	4	21	"	"
R28	CP to 2Q2	"	21	"	"	"	"
R29	CP to 2Q1	"	25	"	"	"	"
R30	CP to 2Q0	"	26	"	"	"	"
S1	CP to 1TC	Propagation delay time (t_{PHL}), C_L = 50 pF	15	"	16	"	"
S2	CP to 1Q3	"	8	5	18	"	"
S3	CP to 1Q2	"	12	"	"	"	"
S4	CP to 1Q1	"	13	"	"	"	"
S5	CP to 1Q0	"	14	"	"	"	"
S6	CP to 2TCA	"	39	4	16	"	"
S7	CP to 2Q7	"	35	5	18	"	"
S8	CP to 2Q6	"	36	"	"	"	"
S9	CP to 2Q5	"	37	"	"	"	"
S10	CP to 2Q4	"	38	"	"	"	"
S11	CP to 2TCB	"	27	4	16	"	"
S12	CP to 2Q3	"	22	5	18	"	"
S13	CP to 2Q2	"	21	"	"	"	"
S14	CP to 2Q1	"	25	"	"	"	"
S15	CP to 2Q0	"	26	"	"	"	"

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
S16	CP to 1TC Propagation delay time (t_{PHL}), C_L = 150 pF	15	4	24	ns	17,18
S17	CP to 1Q3 "	8	5	26	"	"
S18	CP to 1Q2 "	12	"	"	"	"
S19	CP to 1Q1 "	13	"	"	"	"
S20	CP to 1Q0 "	14	"	"	"	"
S21	CP to 2TCA "	39	4	24	"	"
S22	CP to 2Q7 "	35	5	26	"	"
S23	CP to 2Q6 "	36	"	"	"	"
S24	CP to 2Q5 "	37	"	"	"	"
S25	CP to 2Q4 "	38	"	"	"	"
S26	CP to 2TCB "	27	4	24	"	"
S27	CP to 2Q3 "	22	5	26	"	"
S28	CP to 2Q2 "	21	"	"	"	"
S29	CP to 2Q1 "	25	"	"	"	"
S30	CP to 2Q0 "	26	"	"	"	"

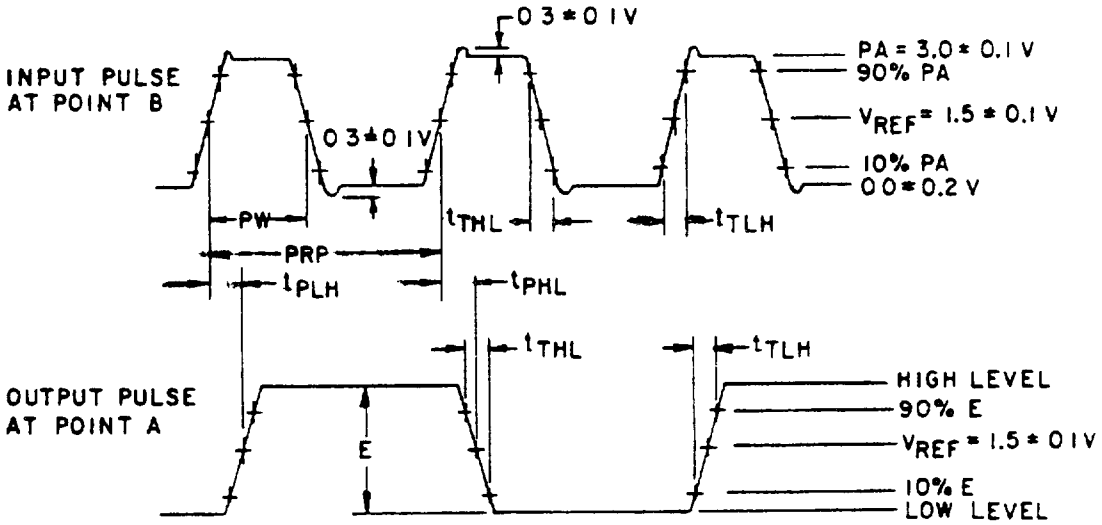
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NOTES.
1. R1 = 51Ω ±5%, 1/4 W.
2. S1 = SPST switch.
3. 93S16 or equivalent = 4 bit counter. Place counter in count mode (PE, CEP, and CET = High level). Trigger item D05 off Q3 (MSB output).
4. After making all circuit connections for a test condition, momentarily close switch S1 for a minimum of 100 ns.
5. All outputs shall be loaded simultaneously as specified on figure 13.

FIGURE 17. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XV.

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NOTES
1. Input pulse parameters
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   PW = 100 ±10 ns.
   PRP = 500 ±50 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pins, between items Y01-1 and Y01-2 and input and output pins, and between load circuit and output pin shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 18. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XV.

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TABLE XVI Propagation delay time (t_{PHL}) test

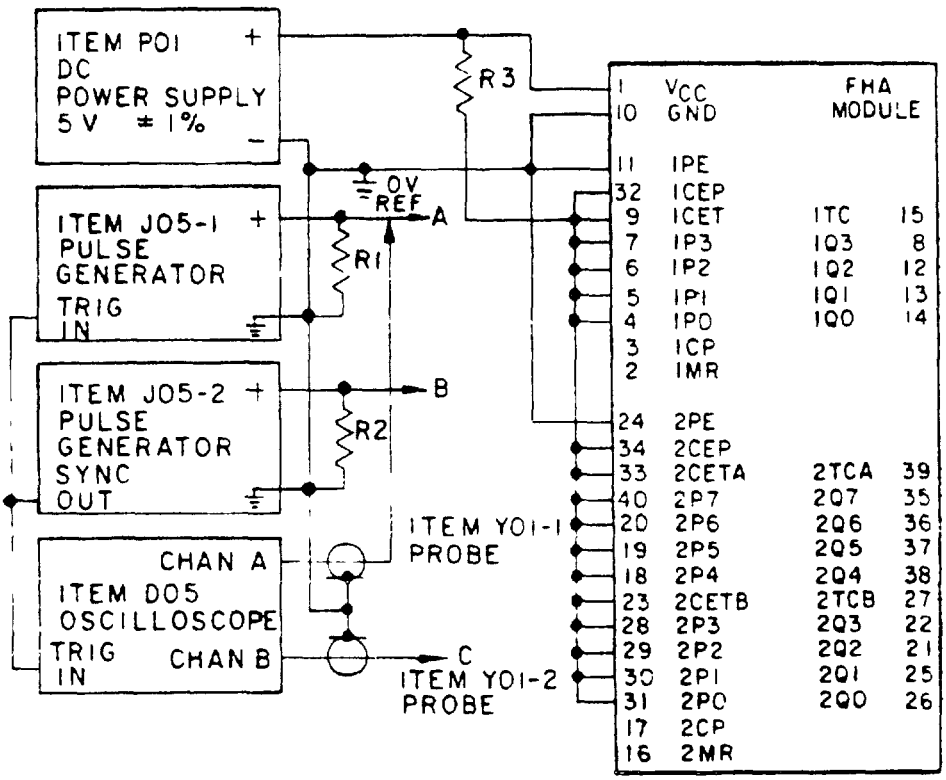
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T1	MR to 1TC Propagation delay time (t_{PHL}), C_L = 50 pF	2	3	15	5	25	ns	19,20
T2	MR to 1Q3 "	"	"	8	"	"	"	"
T3	MR to 1Q2 "	"	"	12	"	"	"	"
T4	MR to 1Q1 "	"	"	13	"	"	"	"
T5	MR to 1Q0 "	"	"	14	"	"	"	"
T6	MR to 2TCA "	16	17	39	"	"	"	"
T7	MR to 2Q7 "	"	"	35	"	"	"	"
T8	MR to 2Q6 "	"	"	36	"	"	"	"
T9	MR to 2Q5 "	"	"	37	"	"	"	"
T10	MR to 2Q4 "	"	"	38	"	"	"	"
T11	MR to 2TCB "	"	"	27	"	"	"	"
T12	MR to 2Q3 "	"	"	22	"	"	"	"
T13	MR to 2Q2 "	"	"	21	"	"	"	"
T14	MR to 2Q1 "	"	"	25	"	"	"	"
T15	MR to 2Q0 "	"	"	26	"	"	"	"
T16	MR to 1TC Propagation delay time (t_{PHL}), C_L = 150 pF	2	3	15	"	33	"	"
T17	MR to 1Q3 "	"	"	8	"	"	"	"
T18	MR to 1Q2 "	"	"	12	"	"	"	"
T19	MR to 1Q1 "	"	"	13	"	"	"	"
T20	MR to 1Q0 "	"	"	14	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T21	MR to 2TCA Propagation delay time (t_{PHL}), C_L = 150 pF	16	17	39	5	33	ns	19,20
T22	MR to 2Q7 "	"	"	35	"	"	"	"
T23	MR to 2Q6 "	"	"	36	"	"	"	"
T24	MR to 2Q5 "	"	"	37	"	"	"	"
T25	MR to 2Q4 "	"	"	38	"	"	"	"
T26	MR to 2TCB "	"	"	27	"	"	"	"
T27	MR to 2Q3 "	"	"	22	"	"	"	"
T28	MR to 2Q2 "	"	"	21	"	"	"	"
T29	MR to 2Q1 "	"	"	25	"	"	"	"
T30	MR to 2Q0 "	"	"	26	"	"	"	"

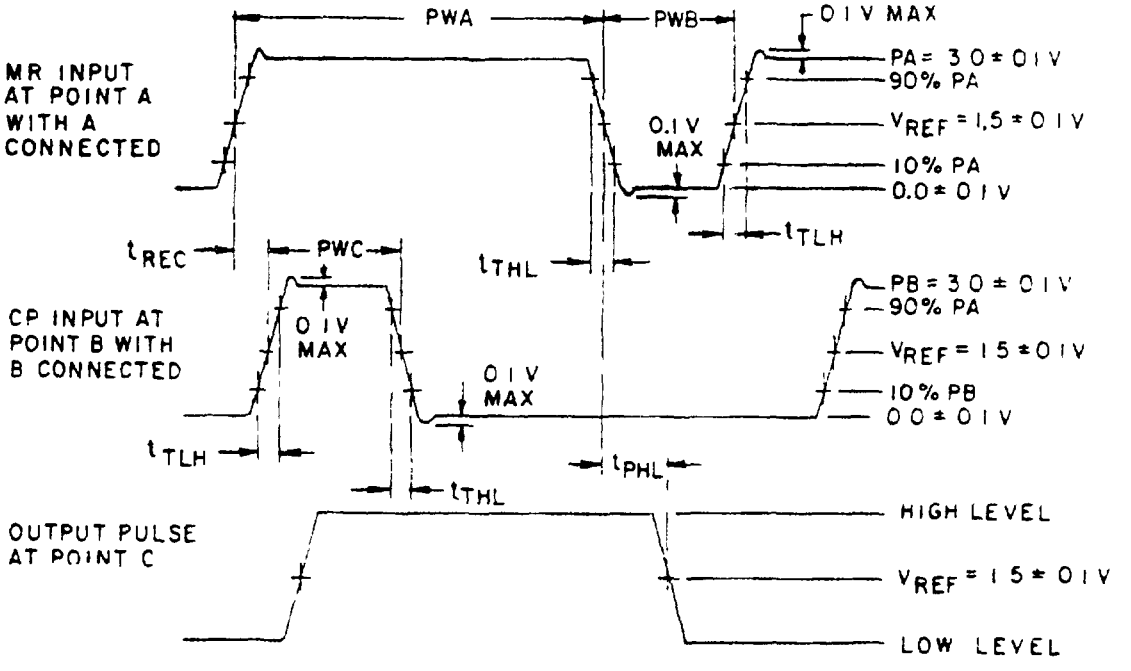
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NOTES
1. R1 = R2 = 51 ±5 , 1/4 w.
2. R3 = 1 k. ±5 , 1/4 w
3. All outputs shall be loaded simultaneously as specified on figure 13.

FIGURE 19 Propagation delay time (t_{PHL}) test circuit for table XVI.

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NOTES
1. Input pulse parameters.
   PWA = 300 ±30 ns.
   PWB = 14 ±1 ns.
   PWC = 100 ±10 ns.
   t_{REC} = 8 ±1 ns.
   $t_{THL} = t_{TLH}$ = 4 ±1 ns.
2. Before making the measurements, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.5 V.
3. The lead length of electrical conductors between resistors R1 and R2 and input pins, between items YO1-1 and YO1-2 and input and output pins, and between load circuit to output pin shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

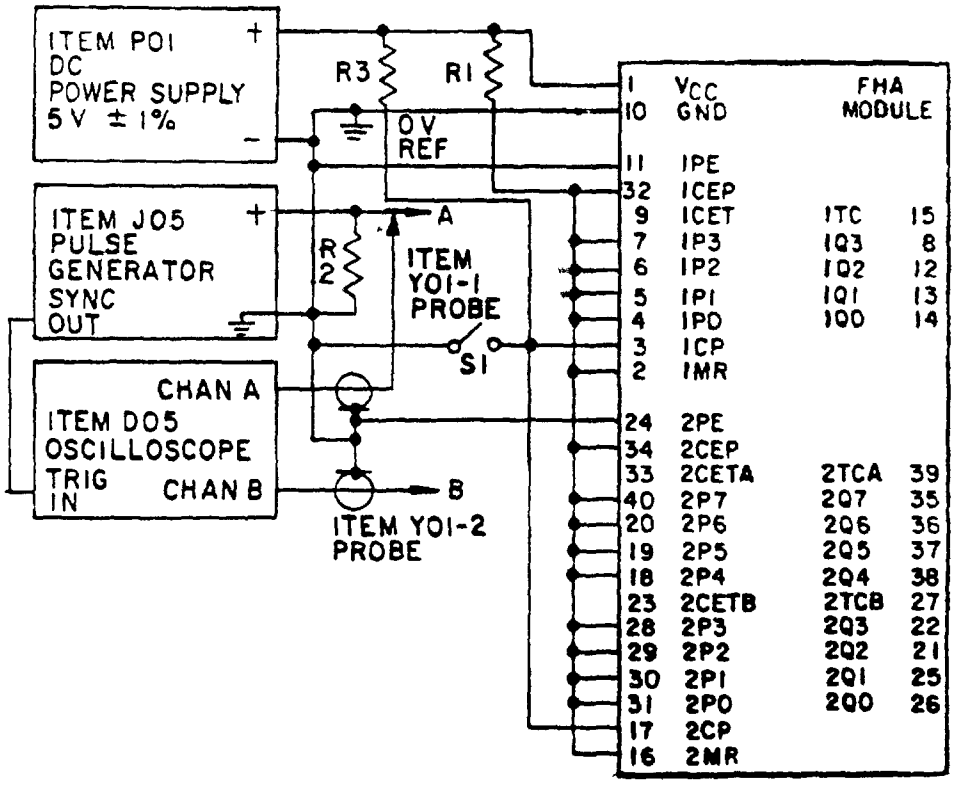
FIGURE 20 Propagation delay time (t_{PHL}) test circuit waveforms for table XVI

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TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U1	1CET to 1TC Propagation delay time (t_{PLH}), C_L = 50 pF	9	15	3	13	ns	21,22
U2	2CETA to 2TCA Propagation delay time (t_{PLH}), C_L = 50 pF	33	39	"	"	"	"
U3	2CETB to 2TCB Propagation delay time (t_{PLH}), C_L = 50 pF	23	27	"	"	"	"
U4	1CET to 1TC Propagation delay time (t_{PLH}), C_L = 150 pF	9	15	"	19	"	"
U5	2CETA to 2TCA Propagation delay time (t_{PLH}), C_L = 150 pF	33	39	"	"	"	"
U6	2CETB to 2TCB Propagation delay time (t_{PLH}), C_L = 150 pF	23	27	"	"	"	"
V1	1CET to 1TC Propagation delay time (t_{PHL}), C_L = 50 pF	9	15	"	14	"	"
V2	2CETA to 2TCA Propagation delay time (t_{PHL}), C_L = 50 pF	33	39	"	"	"	"
V3	2CETB to 2TCB Propagation delay time (t_{PHL}), C_L = 50 pF	23	27	"	"	"	"
V4	1CET to 1TC Propagation delay time (t_{PHL}), C_L = 150 pF	9	15	"	20	"	"
V5	2CETA to 2TCA Propagation delay time (t_{PHL}), C_L = 150 pF	33	39	"	"	"	"
V6	2CETB to 2TCB Propagation delay time (t_{PHL}), C_L = 150 pF	23	27	"	"	"	"

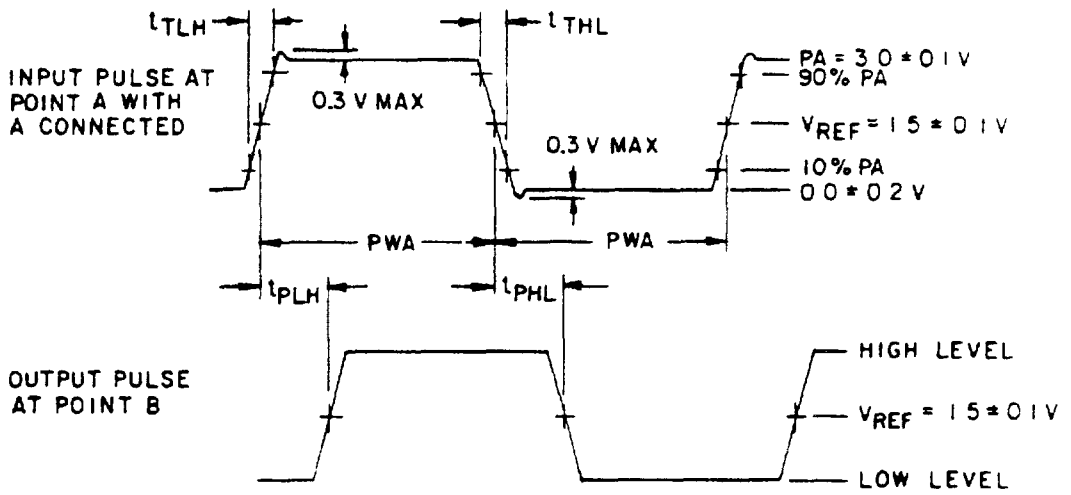
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NOTES
1. R1 = R3 = 1 kΩ ±5%, 1/4 W.
2. R2 = 51Ω ±5%, 1/4 W.
3. S1 = SPST switch.
4. Close switch S1 for a minimum of 100 ns, open switch, and then take measurement.
5. All outputs shall be loaded simultaneously as specified on figure 13.

FIGURE 21. Propagation delay time (t_{PLH} amd t_{PHL}) test circuit for table XVII.

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NOTES
1. Input pulse parameters
   PWA = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 4 ±1 ns.
2. Before making the measurements, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pin, between items YO1-1 and YO1-2 and input and output pins, and between load circuit and output pin shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

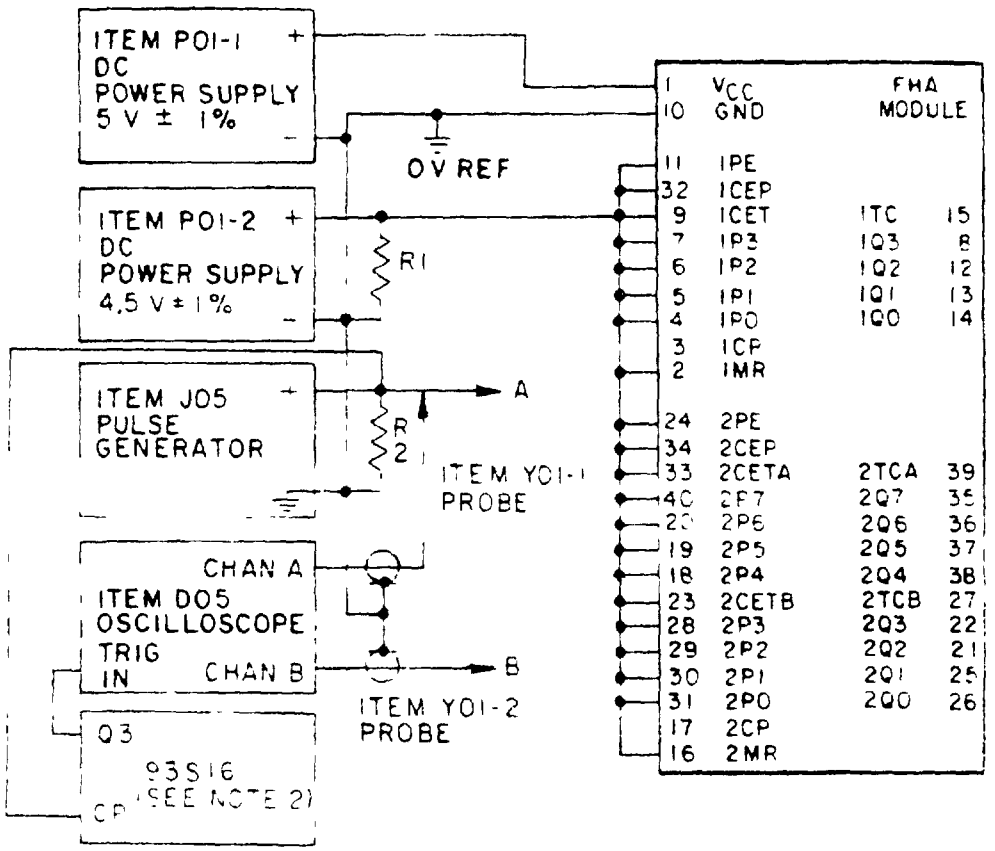
FIGURE 22. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVII

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TABLE XVIII Maximum clock frequency and minimum pulse width test

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W1	1CP Maximum clock frequency and minimum pulse width (PW1)	3	14	21	29	ns	23,24
W2	1CP Maximum clock frequency and minimum pulse width (PW2)	3	14	16	24	"	"
W3	2CP Maximum clock frequency and minimum pulse width (PW1)	17	38	21	29	"	"
W4	2CP Maximum clock frequency and minimum pulse width (PW2)	"	38	16	24	"	"
W5	2CP Maximum clock frequency and minimum pulse width (PW1)	"	26	21	29	"	"
W6	2CP Maximum clock frequency and minimum pulse width (PW2)	"	26	16	24	"	"

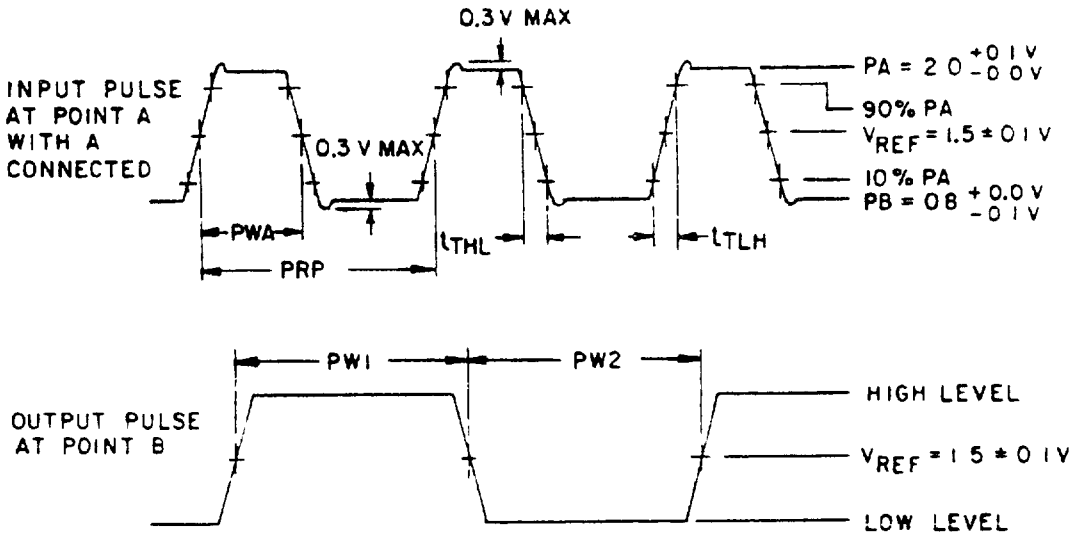
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NOTES
1. R1 - R2 = 51Ω ±5%, 1/4 W.
2. 93S16 or equivalent = 4 bit counter. Place counter in count mode (PE, CEP, and CET = High level). Trigger item D05 off Q3 (MSB output).
3. Load all outputs as specified on figure 13.

FIGURE 23. Maximum clock frequency and minimum pulse width test circuit for table XVIII

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NOTES
1. Input pulse parameters
   PWA = 6 ±1 ns.
   PRP = 22 +2, -0 ns.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
2. Before making the measurements, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.5 V.
3. The lead length of electrical conductors between resistor R2 and input pin, between items YO1-1 and YO1-2 and input and output pins, and between load circuit and output pin shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.
5. Output waveform shall be stable indicator power switching.

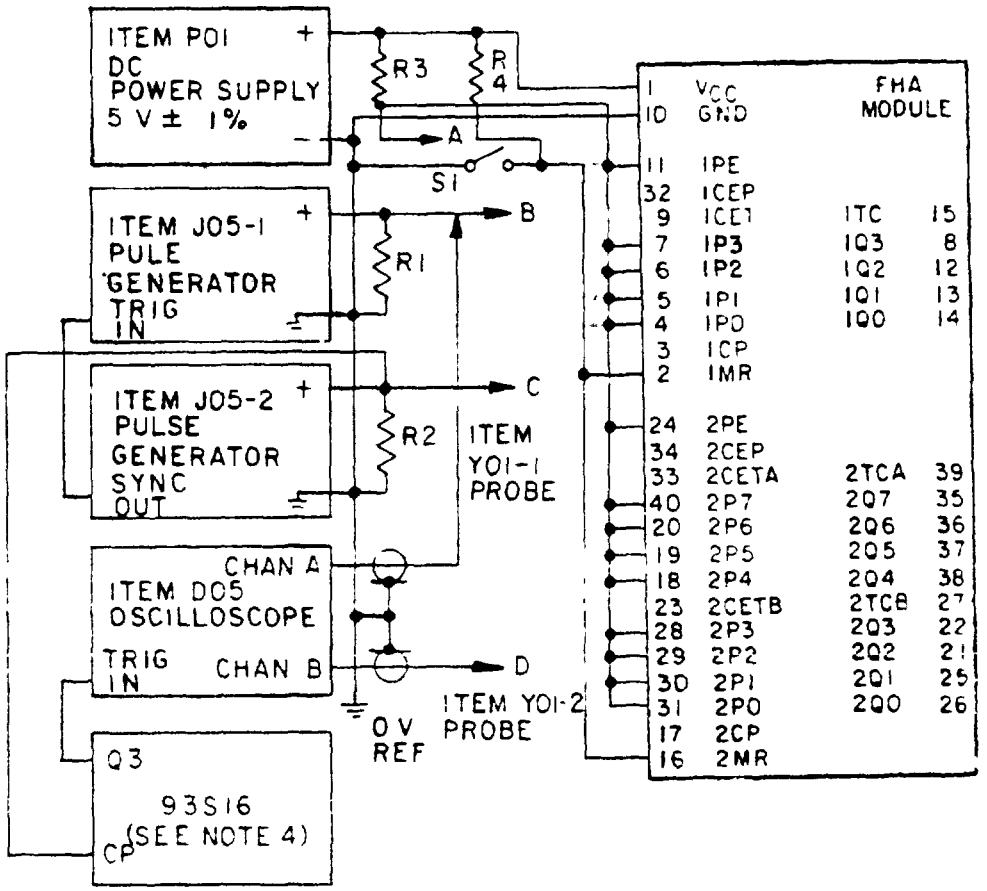
FIGURE 24. Maximum clock frequency and minimum pulse width test circuit waveforms for table XVIII

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TABLE XIX Setup (t_S) and hold (t_H) time check

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X1	1CET Setup time (t_S) check (measuring t_C)	32	9	3	8	1,550	1,650	ns	25,26
X2	2CETA "	34	33	17	35	"	"	"	"
X3	2CETB "	34	23	17	22	"	"	"	"
X4	1CEP "	9	32	3	8	"	"	"	"
X5	2CEP "	33	34	17	35	"	"	"	"
X6	2CEP "	23	34	17	22	"	"	"	"
Y1	1CET Setup time (t_H) check (measuring t_C)	32	9	3	8	"	"	"	"
Y2	2CETA "	34	33	17	35	"	"	"	"
Y3	2CETB "	34	23	17	22	"	"	"	"
Y4	1CEP "	9	32	3	8	"	"	"	"
Y5	2CEP "	33	34	17	35	"	"	"	"
Y6	2CEP "	23	34	17	22	"	"	"	"

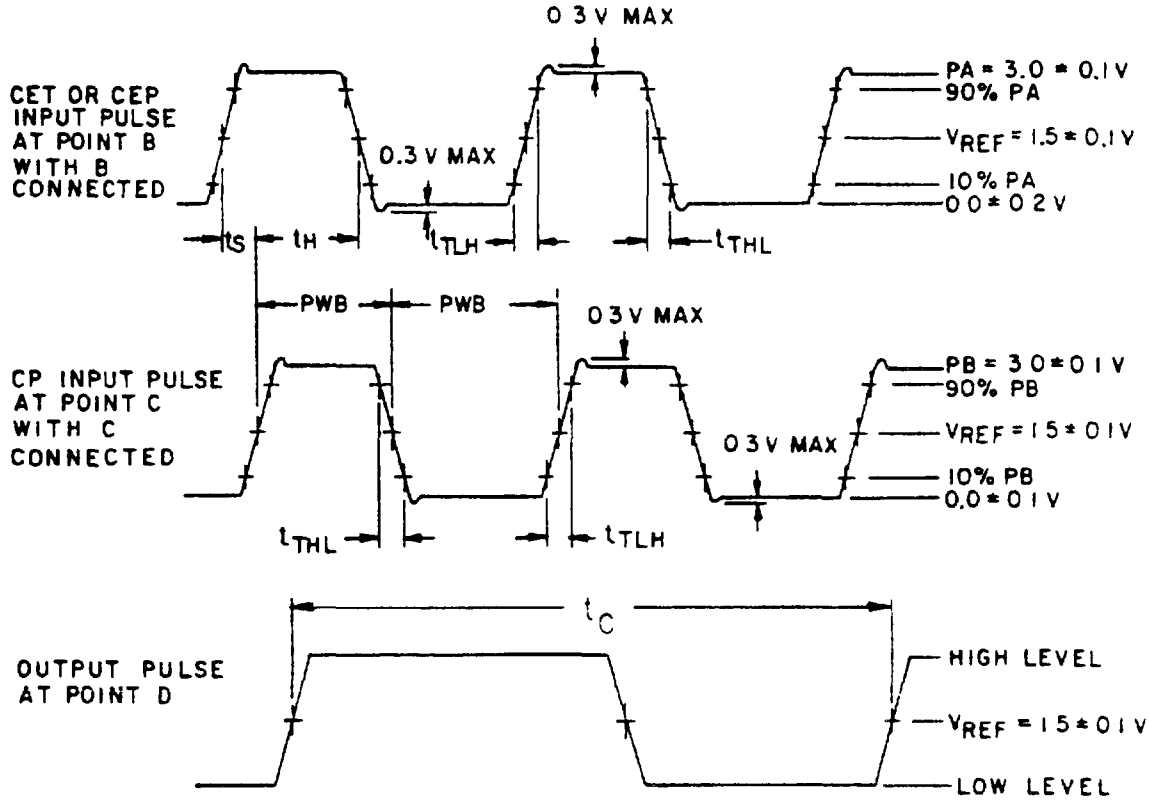
Source: https://assist.dla.mil -- Downloaded: 20260-20T15:52Z
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NOTES
1. R1 = R2 = 51Ω ±5%, 1/4 W.
2. R3 = R4 = 1 kΩ ±5%, 1/4 W.
3. S1 = SPST switch.
4. 93S16 or equivalent = 4 bit counter. Place counter in count mode (PE, CEP, and CET = High level). Trigger item D05 off Q3 (MSB output).
5. After all circuit connections are made, close switch S1 for a minimum of 100 ns then open switch S1 and make measurement.
6. Load all outputs as specified on figure 13.

FIGURE 25. Setup (t_S) and hold (t_H) time test circuit for table XIX.

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NOTES
1. Input pulse parameters for test conditions X1-X6
   t_S = 12 +2, -0 ns.
   t_H = 25 ±3 ns.
   PWB = 50 ±3 ns
   $t_{TLH} = t_{THL}$ = 4 ±1 ns.
2. Input pulse parameters for test conditions Y1-Y6.
   t_S = 25 ±3 ns.
   t_H = 0 +2, -0 ns.
   PWB = 50 ±3 ns.
   $t_{TLH} = t_{THL}$ = 4 ±1 ns.
3. Before making the measurements, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.5 V.
4. The lead length of electrical conductors between resistros R1 and R2 and input pins; between items YO1-1 and YO1-2 and input and output pins, and between load circuit and output pin shall be less than 1 inch (25.4 mm).
5. All voltages are referenced to ground.
6. Output waveform shall be stable indicating power switching.

FIGURE 26. Setup (t_S) and hold (t_H) time check test circuit waveforms for table XIX

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*TABLE XX. Setup and hold time (low-level and high-level) check test.

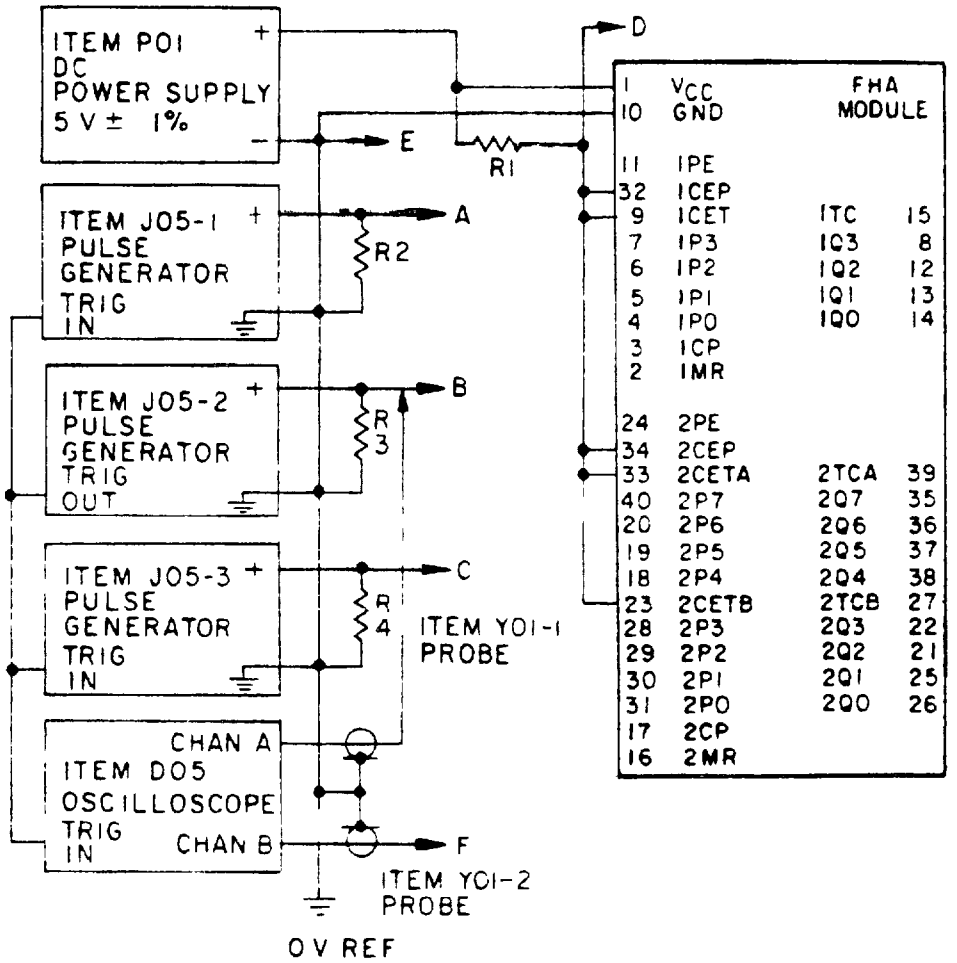
Test condition	Parameter	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
Z1	1PE Setup and hold time check (t_{PLH})	11	3	2	4,5, 6,7	Open	8	4	13	ns	27,29
Z2	2PE "	24	17	16	18,19, 20,40	"	35	"	"	"	"
Z3	3PE "	24	17	16	28,29, 30,31	"	22	"	"	"	"
Z4	1P0 Setup and hold time, (low level), check (t_{PHL})	4	3	Open	2,5, 6,7	11	14	5	18	"	28,30
Z5	1P1 "	5	"	"	2,4, 6,7	"	13	"	"	"	"
Z6	1P2 "	6	"	"	2,4, 5,7	"	12	"	"	"	"
Z7	1P3 "	7	"	"	2,4, 5,6	"	8	"	"	"	"
Z8	2P7 "	40	17	"	16,18, 19,20	24	35	"	"	"	"
Z9	2P6 "	20	"	"	16,18, 19,40	"	36	"	"	"	"
Z10	2P5 "	19	"	"	16,18, 20,40	"	37	"	"	"	"
Z11	2P4 "	18	"	"	16,19, 20,40	"	38	"	"	"	"
Z12	2P3 "	28	"	"	16,29, 30,31	"	22	"	"	"	"
Z13	2P2 "	29	"	"	16,28, 30,31	"	21	"	"	"	"
Z14	2P1 "	30	"	"	16,28, 29,31	"	25	"	"	"	"

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*TABLE XX. Setup and hold time (low-level and high-level) check test - Continued.

Test condition	Parameter	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
Z15	2P0 Setup and hold time, (low level), check (t_{PHL})	31	17	Open	16,28, 29,30	24	26	5	18	ns	28,30
Z16	1P3 Setup and hold time, (high level), check (t_{PLH})	7	3	"	2,4, 5,6	11	8	4	13	"	28,31
Z17	1P2 "	6	"	"	2,4, 5,7	"	12	"	"	"	"
Z18	1P1 "	5	"	"	2,4, 6,7	"	13	"	"	"	"
Z19	1P0 "	4	"	"	2,5, 6,7	"	14	"	"	"	"
Z20	2P7 "	40	17	"	16,18, 19,20	24	35	"	"	"	"
Z21	2P6 "	20	"	"	16,18, 19,40	"	36	"	"	"	"
Z22	2P5 "	19	"	"	16,18, 20,40	"	37	"	"	"	"
Z23	2P4 "	18	"	"	16,19, 20,40	"	38	"	"	"	"
Z24	2P3 "	28	"	"	16,29, 30,31	"	22	"	"	"	"
Z25	2P2 "	29	"	"	16,28, 30,31	"	21	"	"	"	"
Z26	2P1 "	30	"	"	16,28, 29,31	"	25	"	"	"	"
Z27	2P0 "	31	"	"	16,28, 29,30	"	26	"	"	"	"

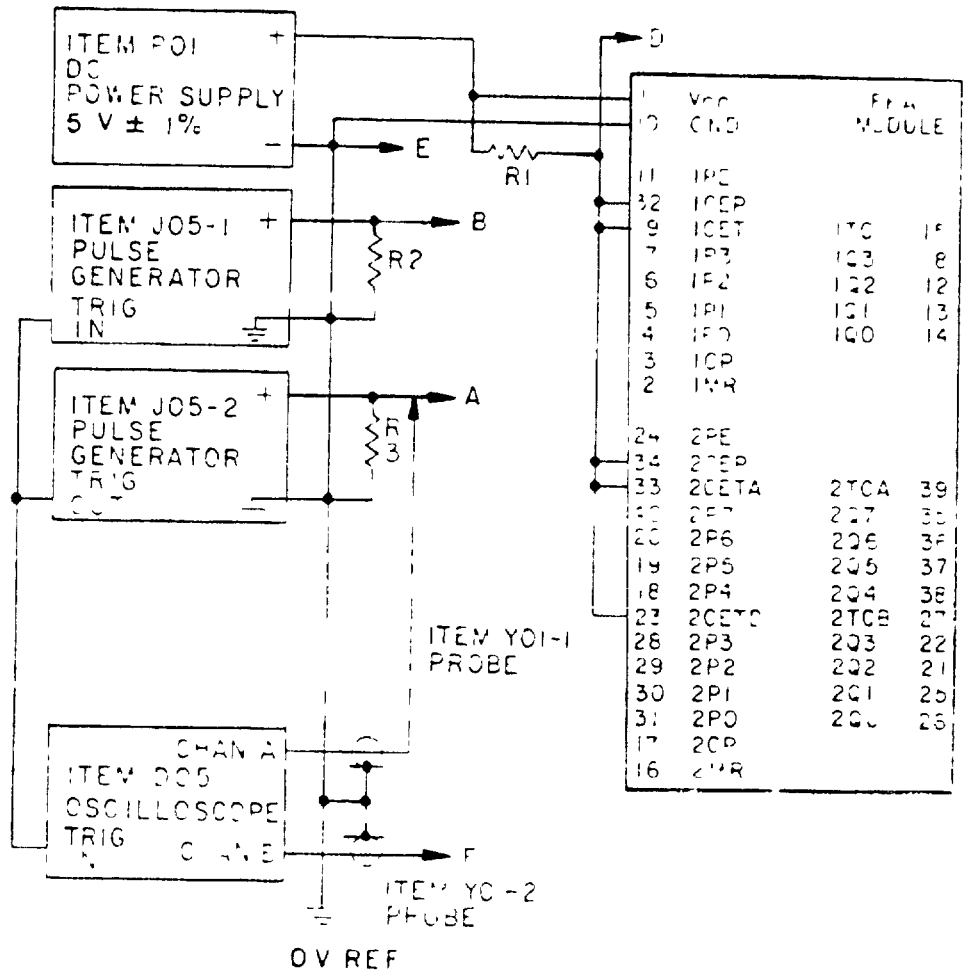
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:52Z
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NOTES
1. R1 = 1 kΩ ±5%, 1/4 W.
2. R2 = R3 = R4 = 51Ω ±5%, 1/4 W.
3. Load all outputs as specified on figure 13.

* FIGURE 27. Setup and hold time check test circuit for table XX.

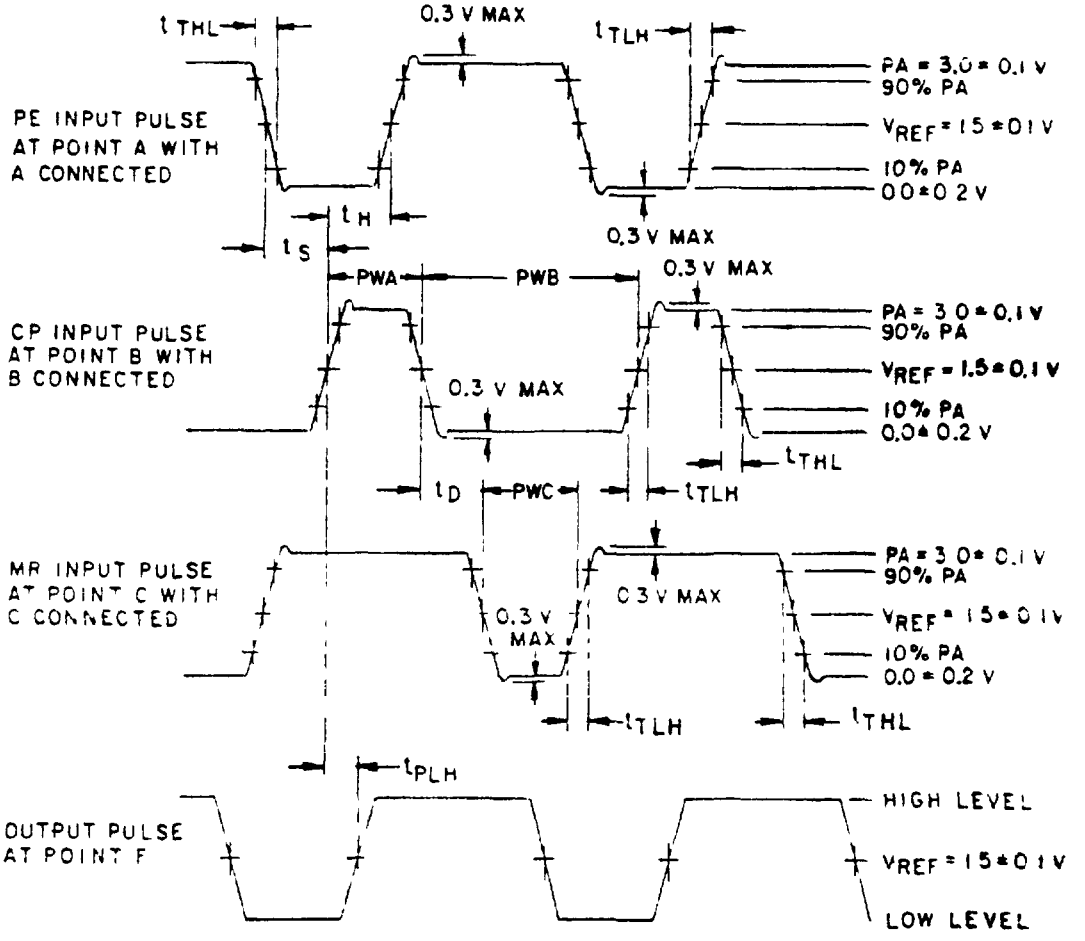
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NOTES
1. R1 = 1 kΩ ±5%, 1/4 W.
2. R2 = R3 = 51Ω ±5%, 1/4 W.
3. Load all outputs as specified on figure 13.

* FIGURE 28 Setup and hold time (low-level and high level) check test circuit for table XV

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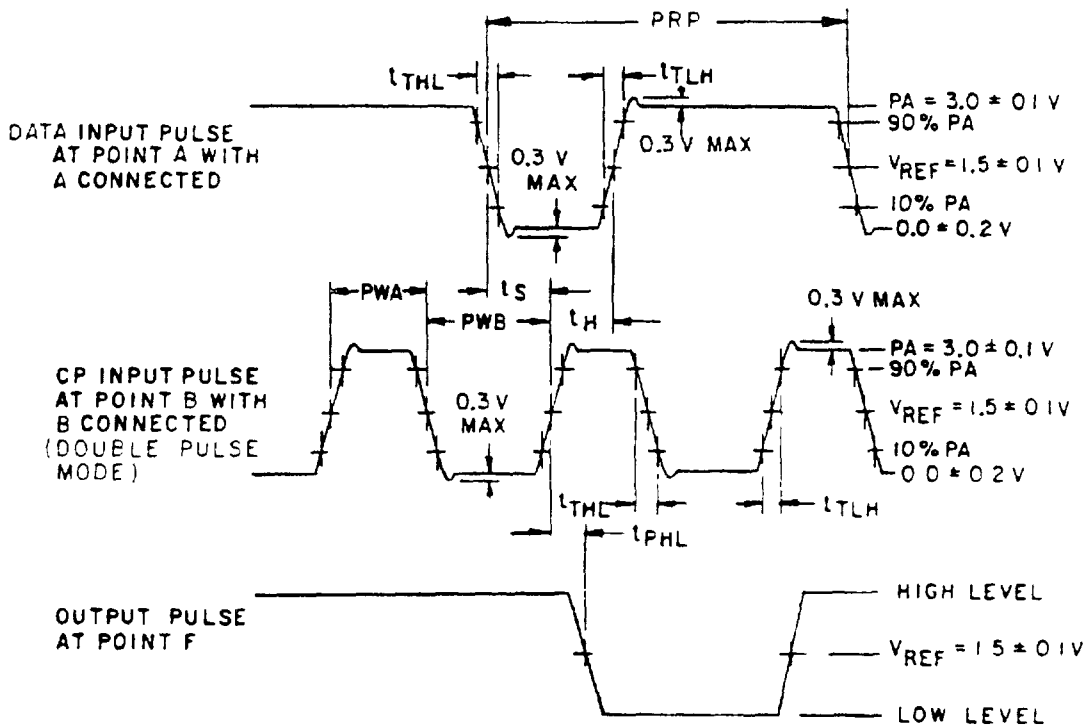


NOTES

1. Input pulse parameters for test conditions Z1-Z3
   PWA = 100 ± 5 ns
   PWB = 200 ±20 ns.
   PWC = 100 ± 5 ns
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns
   t_S = 16 +2, -0 ns.
   t_H = 0 +2, -0 ns.
   t_D = 50 ±5 ns.
2. Before making the measurements, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.5 V.
3. The lead length of electrical conductors between test socket [illegible] and input pins, between items YO1-1 and YO1-2 and input and output [illegible] and between load circuit and output pin shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 29 Setup and hold time check test circuit waveforms for table XX.

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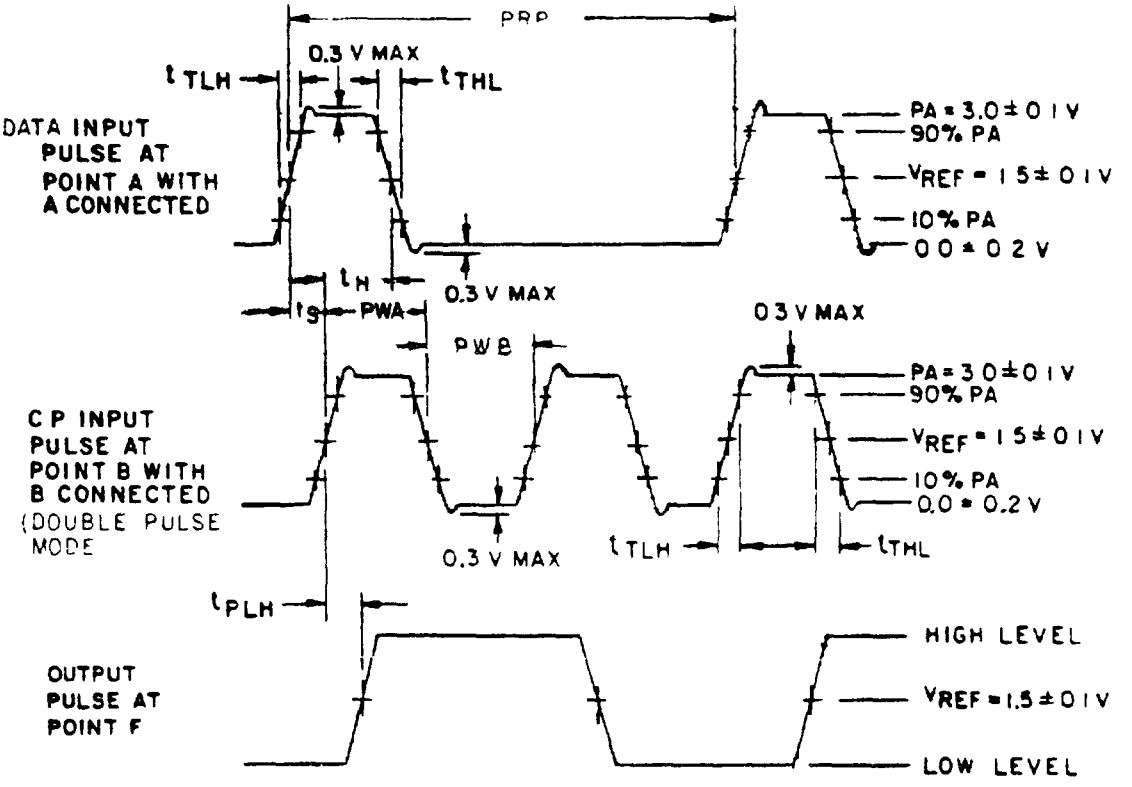


NOTES

1. Input pulse parameters for test conditions 2a-2b
   PWA = 100 ±10 ns
   PWB = 100 ±10 ns
   PRP = 400 ±40 ns
   $t_{TLH} = t_{THL} - 4 -1$ ns.
   t_S = 10 +2, -0 ns
   t_H = 0 +2, -0 ns.
2. Before making the measurements, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.5 V.
3. The lead length of electrical conductors between resistors R2 and R3 and input pins, between items YO1-1 and YO1-2 and input and output pins, and between load circuit and output pin shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 30 Setup and hold time (low-level) check test circuit waveforms for table XX

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NOTES

1. Input pulse parameters for test conditions Z16-Z27
   PWA = 100 ±10 ns.
   PWB = 100 ±10 ns.
   PRP = 400 ±40 ns
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   t_S = 10 +2, -0 ns.
   t_H = 0 +2, -0 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistors R2 and R3 and input pins, between items YO1-1 and YO1-2 and input and output pins, and between load circuit and output pin shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 31. Setup and hold time (high-level) check test circuit waveforms for table XX

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TABLE XXI Output crosstalk (low-level, duration, and high-level) test

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
AA1	1TC Output crosstalk, low level	23,33	9	40,31,30,29, 28,20,19,18, 7,6,5,4	3,17	15	0	1,250	mV	32,33
AA2	1Q3 "	9,23,33	7	40,31,30,29, 28,20,19,18, 6,5,4	"	8	"	"	"	"
AA3	1Q2 "	"	6	40,31,30,29, 28,20,19,18, 7,5,4	"	12	"	"	"	"
AA4	1Q1 "	"	5	40,31,30,29, 28,20,19,18, 7,6,4	"	13	"	"	"	"
AA5	1Q0 "	"	4	40,31,30,29, 28,20,19,18, 7,6,5	"	14	"	"	"	"
AA6	2TCA "	9,23	33	40,31,30,29, 28,20,19,18, 7,6,5,4	"	39	"	"	"	"
AA7	2Q7 "	9,23,33	40	31,30,29,28, 20,19,18,7,6, 5,4	"	35	"	"	"	"
AA8	2Q6 "	"	20	40,31,30,29, 28,19,18,7,6, 5,4	"	36	"	"	"	"
AA9	2Q5 "	"	19	40,31,30,29, 28,20,18,7,6, 5,4	"	37	"	"	"	"
AA10	2Q4 "	"	18	40,31,30,29, 28,20,19,7,6, 5,4	"	38	"	"	"	"
AA11	2TCB "	9,33	23	40,31,30,29, 28,20,19,18, 7,6,5,4	"	27	"	"	"	"

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TABLE XXI. Output crosstalk (low-level, duration, and high-level) test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect F to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
AA12	2Q3	Output crosstalk, low level	9,23,33	28	40,31,30,29, 20,19,18,7,6, 5,4	3,17	22	0	1,250	mV	32,33
AA13	2Q2	"	"	29	40,31,30,28, 20,19,18,7,6, 5,4	"	21	"	"	"	"
AA14	2Q1	"	"	30	40,31,29,28, 20,19,18,7,6, 5,4	"	25	"	"	"	"
AA15	2Q0	"	"	31	40,30,29,28, 20,19,18,7,6, 5,4	"	26	"	"	"	"
AA16	1TC	Output crosstalk, duration	23,33	9	40,31,30,29, 28,20,19,18, 7,6,5,4	"	15	"	500	"	32,34
AA17	1Q3	"	9,23 33	7	40,31,30,29, 28,20,19,18, 6,5,4	"	8	"	"	"	"
AA18	1Q2	"	"	6	40,31,30,29, 28,20,19,18, 7,5,4	"	12	"	"	"	"
AA19	1Q1	"	"	5	40,31,30,29, 28,20,19,18, 7,6,4	"	13	"	"	"	"
AA20	1Q0	"	'	4	40,31,30,29, 28,20,19,18, 7,6,5	"	14	"	"	"	"
AA21	2TCA	"	9,23	33	40,31,30,29, 28,20,19,18, 7,6,5,4	"	39	"	"	"	"
AA22	2Q7	"	9,23,33	40	31,30,29,28, 20,19,18,7,6, 5,4	"	35	"	"	"	"

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TABLE XXI. Output crosstalk (low-level, duration, and high-level) test Continued

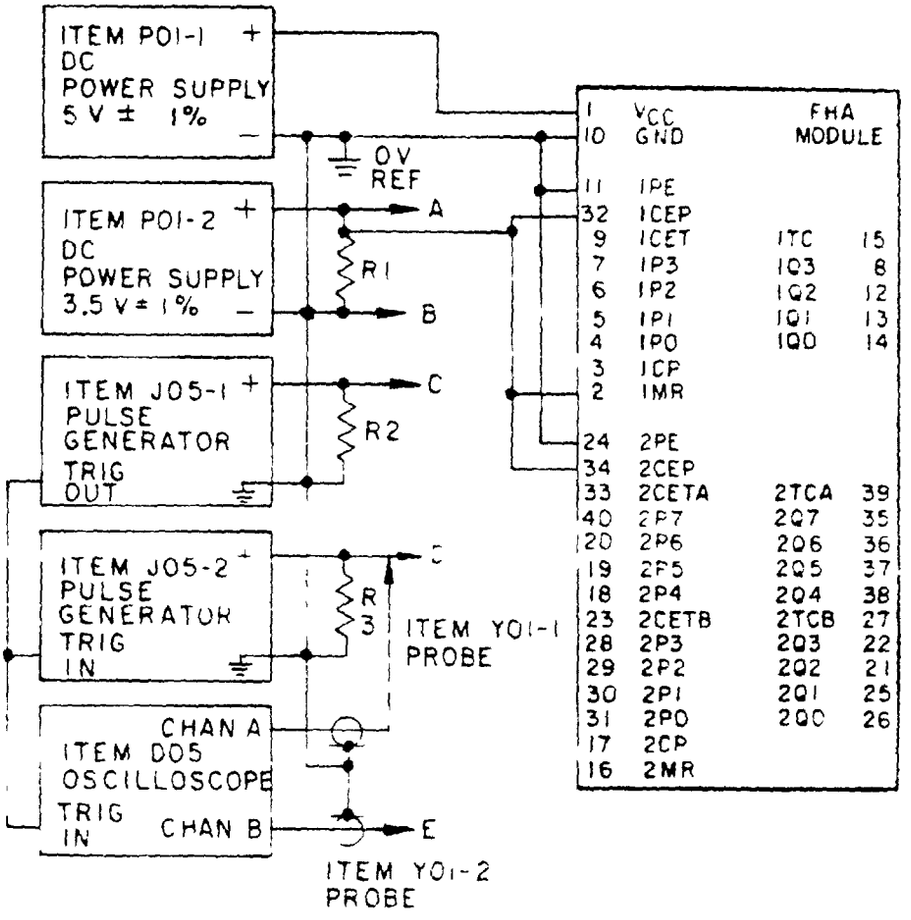
Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
AA23	2Q6	Output crosstalk, duration	9,23,33	20	40,31,30,29, 28,17,18,7,5, 5,4	3,17	36	0	500	mV	32,34
AA24	2Q5	"	"	19	40,31,30,29, 28,20,18,7,6, 5,4	"	37	"	"	"	"
AA25	2Q4	"	"	18	40,31,30,29, 28,20,19,7,6, 5,4	"	38	"	"	"	"
AA26	2TCB	"	9,33	23	40,31,30,29, 28,20,19,18, 7,6,5,4	"	27	"	"	"	"
AA27	2Q3	"	9,23,33	28	40,31,30,29, 20,19,18,7,6, 5,4	"	22	"	"	"	"
AA28	2Q2	"	"	29	40,31,30,28 20,19,18,7,6, 5,4	"	21	"	"	"	"
AA29	2Q1	"	"	30	40,31,29,28, 20,19,18,7,6, 5,4	"	25	"	"	"	"
AA30	2Q0	"	"	31	40,30,29,28, 20,19,18,7,6, 5,4	"	26	"	"	"	"
BB1	1Q3	Output crosstalk, high level	7,9, 23,33	Open	40,31,30,29, 28,20,19,18, 6,5,4	"	8		400	"	32,33
BB2	1Q2	"	23,33	"	40,31,30,29, 28,20,19,18, 7,5,4	"	12	"	"	"	"
BB3	1Q1	"	5,9, 23,33	"	40,31,30,29, 28,20,19,18, 7,6,4	"	13	"	"	"	"

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TABLE XXI. Output crosstalk (low-level, duration, and high-level) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
BB4	1Q0	Output crosstalk, high level	4,9, 23,33	Open	40,31,30,29, 28,20,19,18, 7,6,5	3,17	14	0	400	mV	32,33
BB5	2Q7	"	9,23, 33,40	"	31,30,29,28, 20,19,18,7,6, 5,4	"	35	"	"	"	"
BB6	2Q6	"	9,20, 23,33	"	40,31,30,29, 28,19,18,7,6, 5,4	"	36	"	"	"	"
BB7	2Q5	"	9,19, 23,33	"	40,31,30,29, 28,20,18,7,6, 5,4	"	37	"	"	"	"
BB8	2Q4	"	9,18, 23,33	"	40,31,30,29, 28,20,19,7,6, 5,4	"	38	"	"	"	"
BB9	2Q3	"	9,23, 28,33	"	40,31,30,29, 20,19, 8,7,6, 5,4	"	22	"	"	"	"
BB10	2Q2	"	9,23, 29,33	"	40,31,30,28, 20,19,18,7,6, 5,4	"	21	"	"	"	"
BB11	2Q1	"	9,23, 30,33	"	40,31,29,28, 20,19,18,7,6, 5,4	"	25	"	"	"	"
BB12	2Q0	"	9,23, 31,33	"	40,30,29,28, 20,19,18,7 6, 5,4	"	26	"	"	"	"

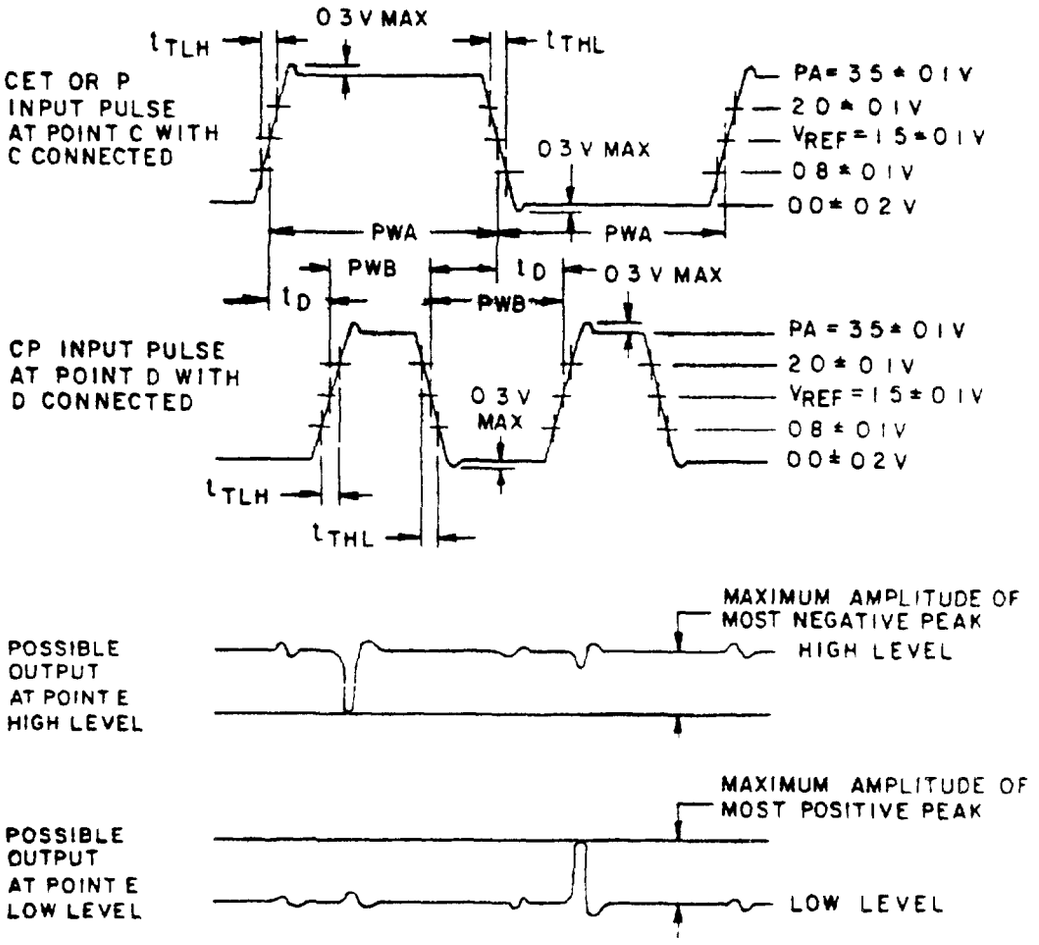
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:52Z
Check the source to verify that this is the current version before use.



NOTES
1 R1 - R3 = 51 ±5 %, 1/4 W.
2 Load all outputs with load as specified on figure 13 with the exception that capacitor C_L is 50 ±5 pF.

FIGURE 32. Output crosstalk (low-level, duration, and high-level) test circuit for table XXI.

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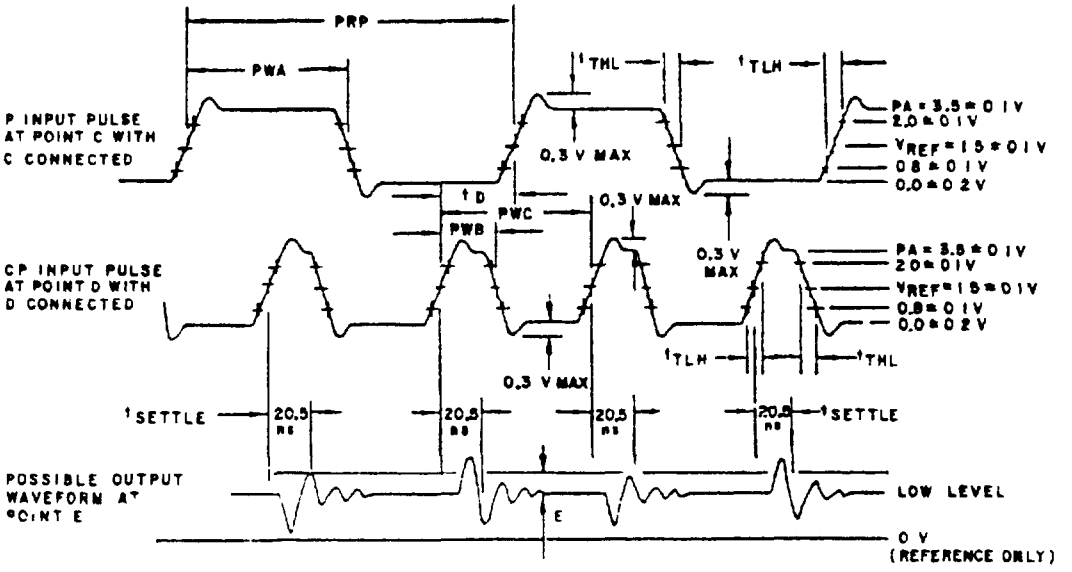


NOTES

1. Input pulse parameters
   PWA = 200 ±20 ns.
   PWB = 100 ±10 ns.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
   t_D = 25 ±3 ns.
2. Before making the measurements, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.5 V.
3. The lead length of electrical conductors between resistors R2 and R3 and input pins, between liters V01-1 and V01-2 and input and output pins, and between load circuit and output pin shall be less than 1 inch (25.4 mm).

FIGURE 33 Output crosstalk (low-level and high-level) test circuit waveforms for table XXI

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NOTES

1. P input pulse parameters
   PWA = 200 ±20 ns.
   PRP = 400 +40 ns
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin,.
2. CP input pulse parameters
   PWB = 100 ±10 ns.
   PWC = 200 +20 ns.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
3. Item J05-2 is operated in the double pulse mode using item J05. Adjust the delay and pulse width controls of items J05-1 and J05-2 to obtain the required P and CP relationships with t_D = 25 ±3 ns
4. E is the maximum amplitude of the most positive peak measured after a settle time of 20.5 ns has elapsed after any positive transition of the CP pulse and before the next positive transition of the CP pulse.
5. The lead length of electrical conductors between resistors R2 and R3 and input pins, between items Y01-1 and Y01-2 and input and output pins, and between load circuit and output pin shall be less than 1 inch (25.4 mm).

* FIGURE 34 Output crosstalk (duration) test circuit waveforms for table XXI

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TABLE XXII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
CC	Power supply transient amplitude	0	400	mV	35,36

TABLE XXIII. Power supply filter capacitance test.

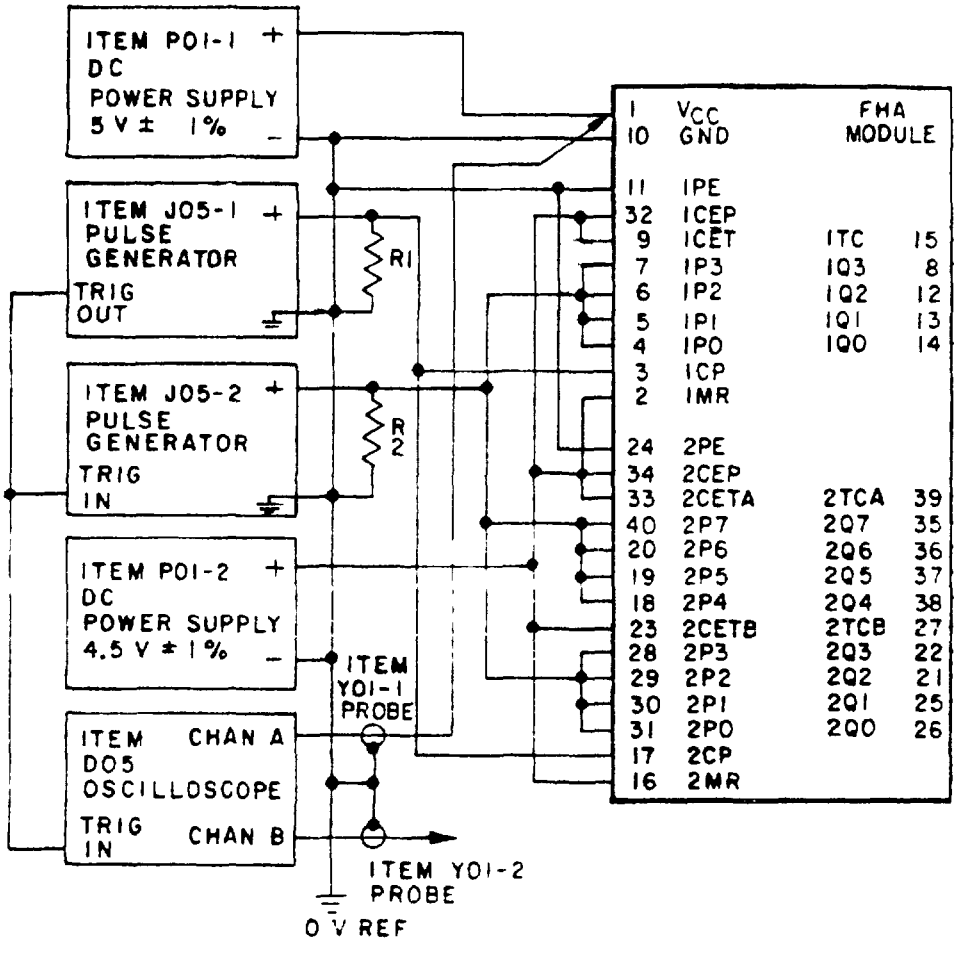
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
DD	Power supply filter capacitance	Open	1	1.6	3.0	µF	37

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TABLE XXIV Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
EE1	1PE Input capacitance	11	Open	5	20	pF	37
EE2	1CEP "	32	"	"	20	"	"
EE3	1CET "	9	"	"	40	"	"
EE4	1P3 "	7	"	"	30	"	"
EE5	1P2 "	6	"	"	"	"	"
EE6	1P1 "	5	"	"	"	"	"
EE7	1P0 "	4	"	"	"	"	"
EE8	1CP "	3	"	"	"	"	"
EE9	1MR "	2	"	"	"	"	"
EE10	2PE "	24	"	"	60	"	"
EE11	2CEP "	34	"	"	60	"	"
EE12	2CETA "	33	"	"	40	"	"
EE13	2P7 "	40	"	"	30	"	"
EE14	2P6 "	20	"	"	"	"	"
EE15	2P5 "	19	"	"	"	"	"
EE16	2P4 "	18	"	"	"	"	"
EE17	2CETB "	23	"	"	40	"	"
EE18	2P3 "	28	"	"	30	"	"
EE19	2P2 "	29	"	"	"	"	"
EE20	2P1 "	30	"	"	"	"	"
EE21	2P0 "	31	"	"	"	"	"
EE22	2CP "	17	"	"	60	"	"
EE23	2MR "	16	"	"	60	"	"

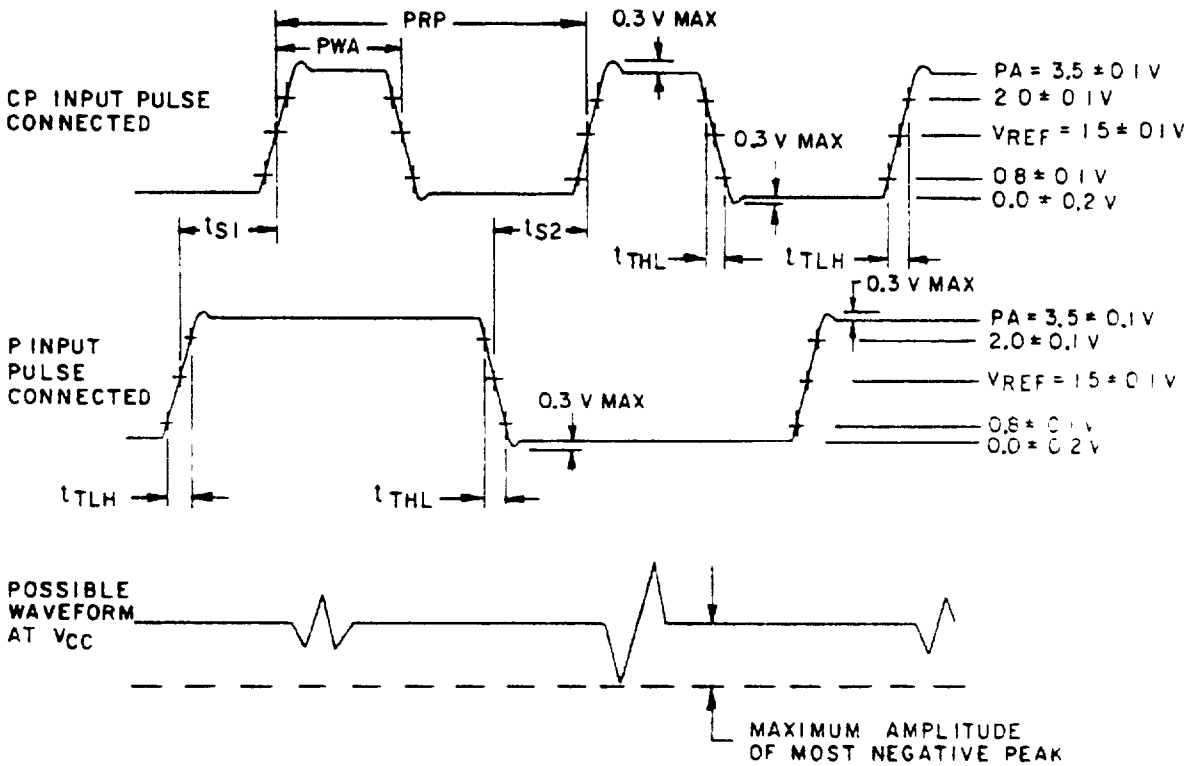
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NOTE R1 = R2 = 51Ω ±5%, 1/4 W

FIGURE 35 Power supply transient amplitude test circuit for table XXII.

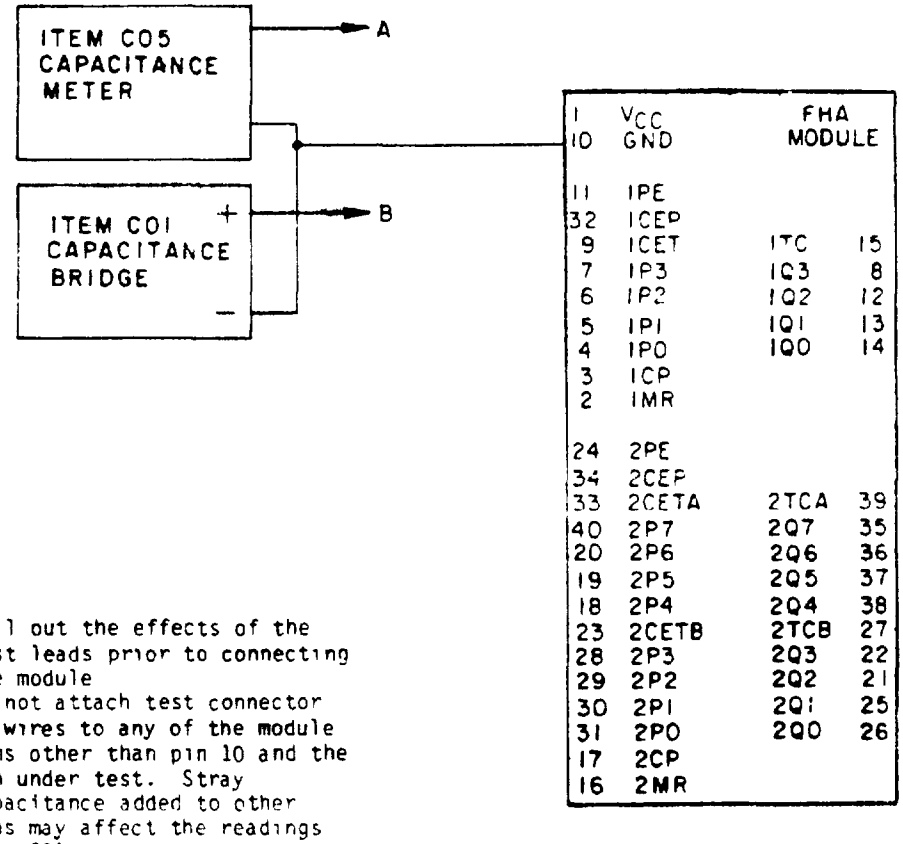
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NOTE Input pulse parameters
PWA = 250 ±25 ns
PRP = 500 ±50 ns
$t_{S1} = t_{S2} = 50 \pm 5$ ns
$t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin)

FIGURE 36 Power supply transient amplitude test circuit waveforms for table XXII.

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NOTES
1 Null out the effects of the test leads prior to connecting the module
2 Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings
3 Item C01 setup
Frequency = 120 Hz.
V_{pk} = 0.3 V maximum

* FIGURE 37. Power supply filter and input capacitance test circuit for tables XXIII and XXIV

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